SUBSET CONSTRUCTION ALGORITHM:

Converting NFA to DFA

SUBSET CONSTRUCTION ALGORITHM:

1. Create the start state of the DFA by taking the **Є-closure** of the start state of the NFA.
2. Perform the following for the new DFA state:
   For each possible input symbol:
   1. Apply move to the newly-created state and the input symbol; this will return a set of states.
   2. Apply the Є -closure to this set of states, possibly resulting in a new set.
3. This set of NFA states will be a single state in the DFA.
4. Each time we generate a new DFA state, we must apply step 2 to it. The process is complete when applying step 2 does not yield any new states.
5. The finish states of the DFA are those which contain any of the finish states of the NFA.

Example: If A, B and C are states, move({A,B,C},`a') = Move (A, 'a') move(B, 'a') move(C, 'a').

Algorithm: Convert NFA to DFA

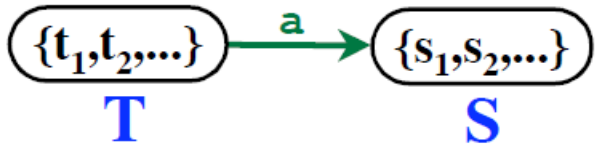
```
S_DFA = {}
Add ε-Closure(s_0) to S_DFA as the start state
Set the only state in S_DFA to "unmarked"
while S_DFA contains an unmarked state do
  Let T be that unmarked state
  Mark T
  for each a in Σ do
    S = ε-Closure(Move_NFA(T,a))
    if S is not in S_DFA already then
      Add S to S_DFA (as an "unmarked" state)
    endIf
    Set Move_DFA(T,a) to S

  endFor
endWhile
for each S in S_DFA do
  if any s∈S is a final state in the NFA then
    Mark S an a final state in the DFA
  endIf
endFor
```

A set of NFA states

Everywhere you could possibly get to on an a

i.e, add an edge to the DFA...



NFA to DFA

Input: A NFA

S = States = $\{ s_0, s_1, \ldots, s_N \} = S_{NFA}$

δ = Move function = Move_{NFA}

Move'(S, a) → Set of states

Output: A DFA

S = States = $\{?, ?, \ldots, ?\} = S_{DFA}$

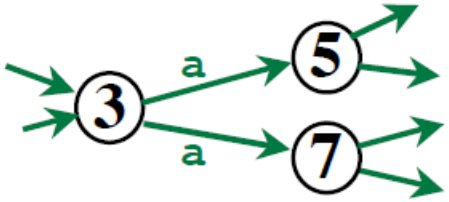
δ = Move function = Move_{DFA}

Move(s, a) → Single state from S_{DFA}

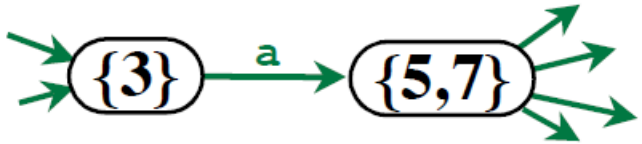
Main Idea:

Each state in S_{DFA} will be a set of states from the NFA

$S_{DFA} = \{ \{\ldots\}, \{\ldots\}, \ldots, \{\ldots\} \}$

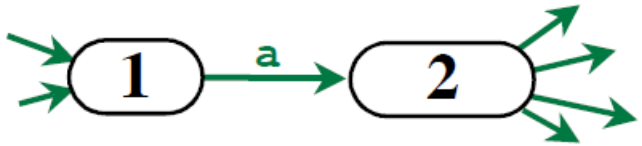


NFA



DFA

(The names of the states is arbitrary and can be changed later, if desired.)



Algorithm: Convert NFA to DFA

We'll use...

$\mathtt{Move}_{NFA}(S,a)$	the transition function from NFA
ε-Closure(s)	where **s** is a single state from NFA
ε-Closure(S)	where **S** is a set of states from NFA

We'll construct...

S_{DFA} the set of states in the DFA

Initially, we'll set S_{DFA} to {}

Add **X** to S_{DFA} where **X** is some *set of* NFA states

Example: "Add {3,5,7} to S_{DFA}"

We'll "mark" some of the states in the DFA.

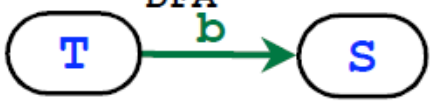
Marked = *"We've done this one"* (√)

Unmarked = *"Still need to do this one"*

$\mathtt{Move}_{DFA}(T,b)$ The transition function from DFA

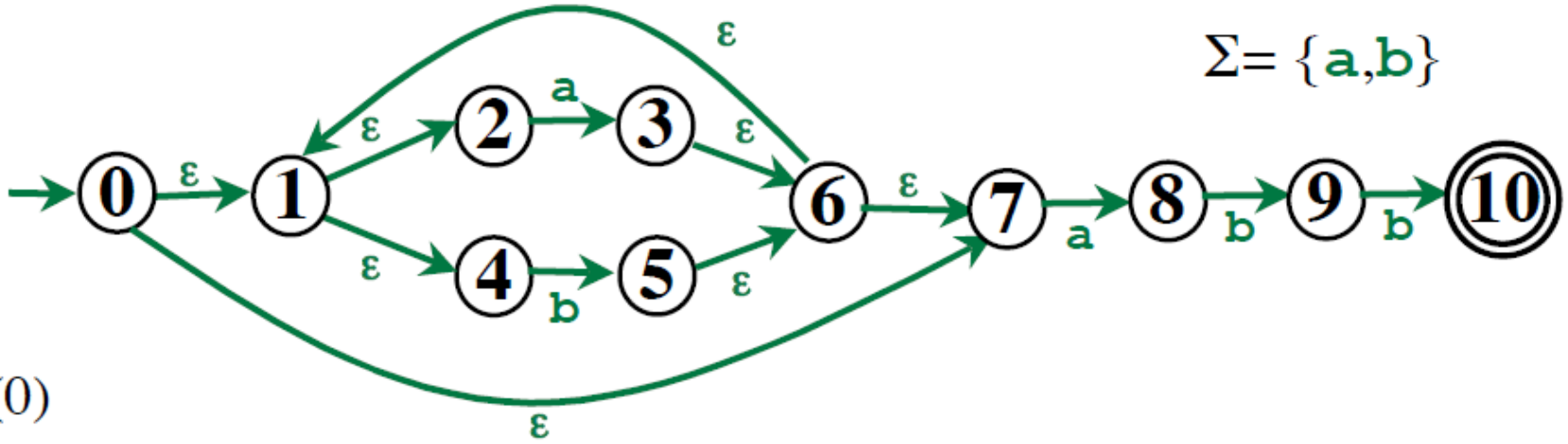
To add an edge to the growing DFA...

Set $\mathtt{Move}_{DFA}(T,b)$ to S



...where **S** and **T** are sets of NFA states

Example



Σ= {a,b}

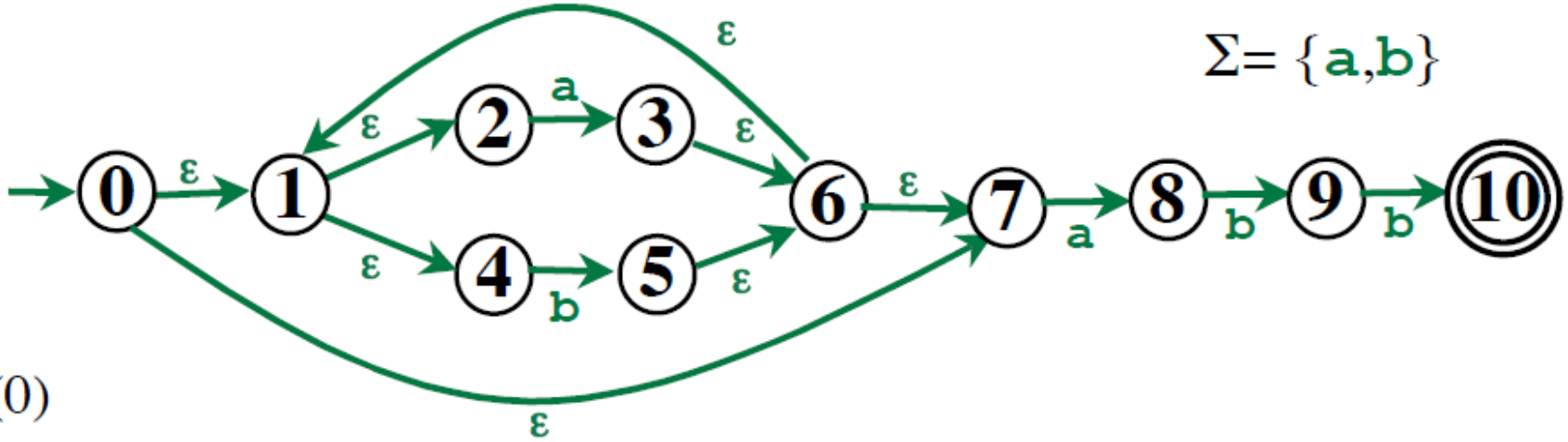
Start state:

ε-Closure (0)

=

Epsilon (∈) – closure : Epsilon closure for a given state X is a set of states which can be reached from the states X with only (null) or ε moves including the state X itself.

Example



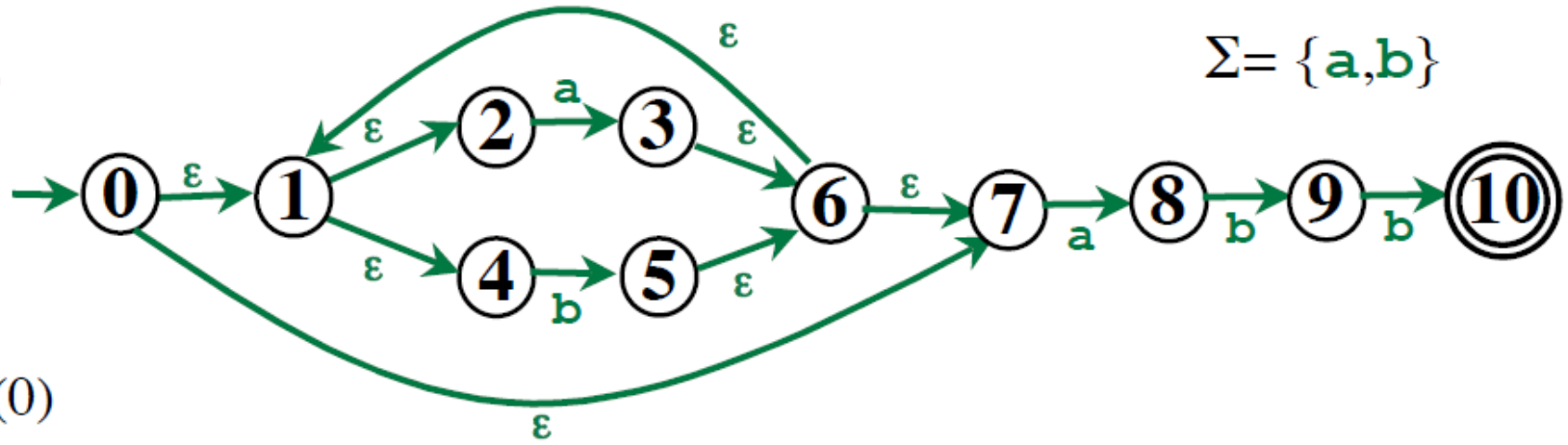
$\Sigma = \{a,b\}$

Start state:

ε-Closure (0)

= {0, 1, 2, 4, 7}

Example



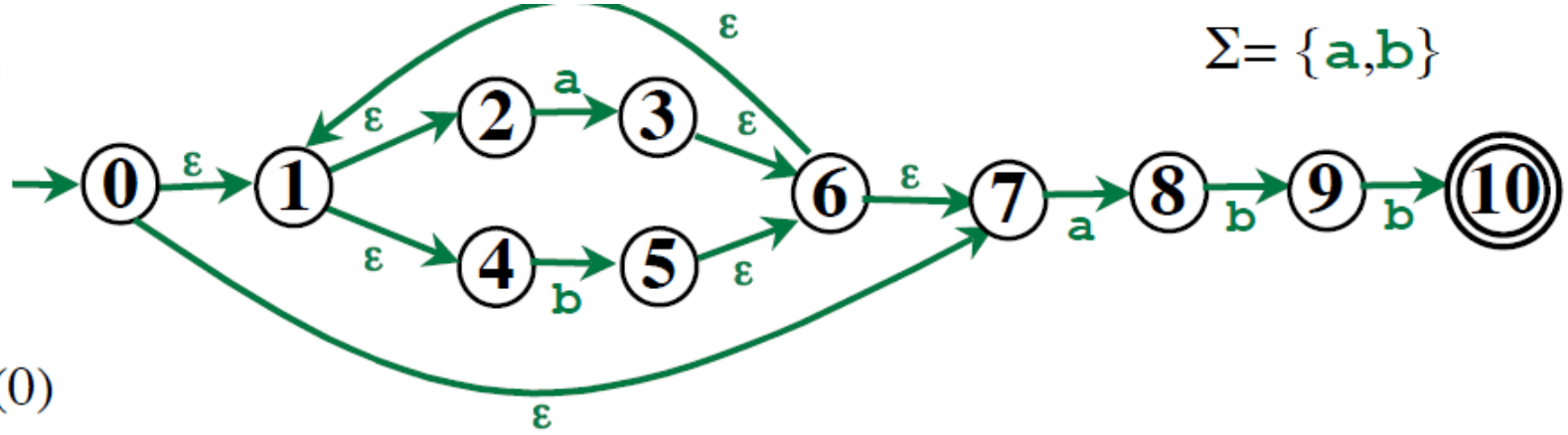
$\Sigma = \{a,b\}$

Start state:

ε-Closure (0)

= {0, 1, 2, 4, 7} = A

Example



$\Sigma = \{a,b\}$

Start state:

ε-Closure (0)

= {0, 1, 2, 4, 7} = A

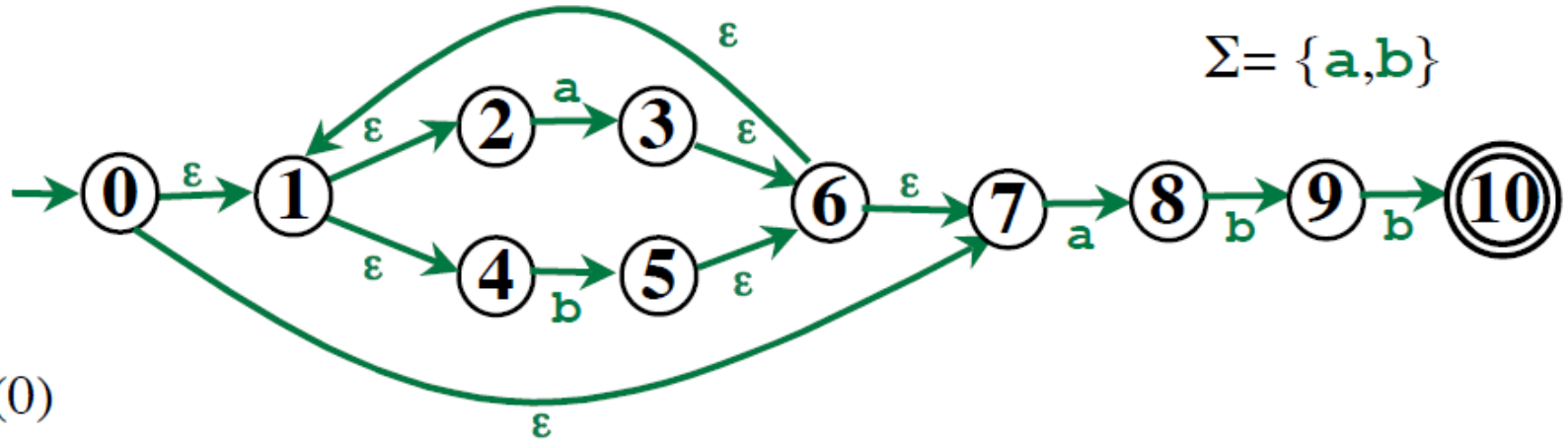
$\text{Move}_{\text{DFA}}(A,a)$

=

$\text{Move}_{\text{DFA}}(A,b)$

=

Example



$\Sigma = \{a,b\}$

Start state:

ε-Closure (0)

= {0, 1, 2, 4, 7} = A

$Move_{DFA}(A,a)$

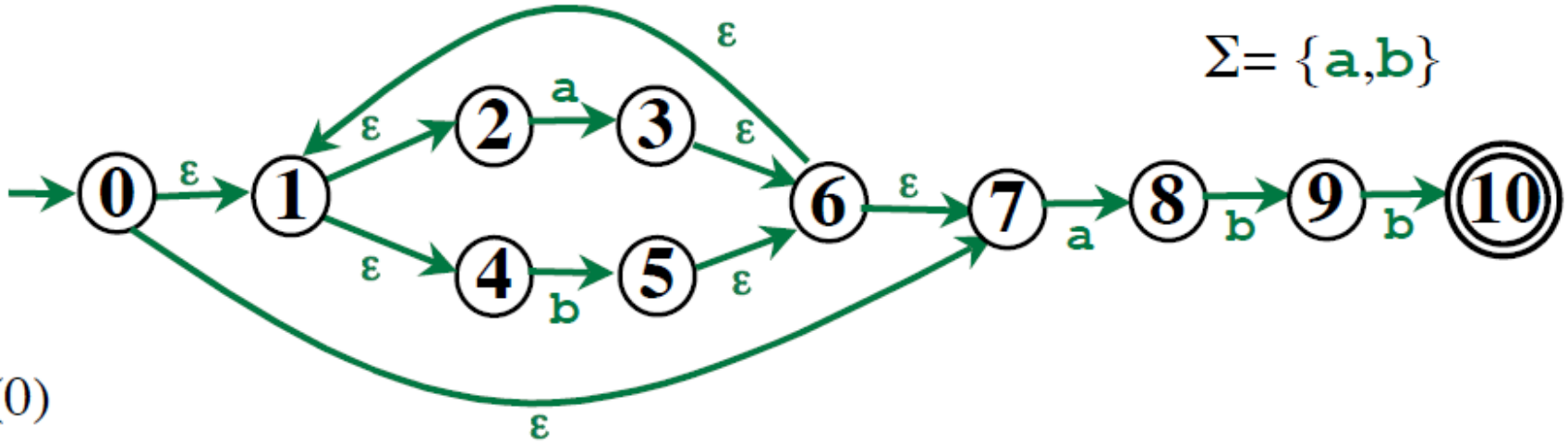
= ε-Closure ($Move_{NFA}(A,a)$)

=

$Move_{DFA}(A,b)$

=

Example



$\Sigma = \{a,b\}$

Start state:

ε-Closure (0)

= {0, 1, 2, 4, 7} = A

$Move_{DFA}(A,a)$

= ε-Closure ($Move_{NFA}(A,a)$)

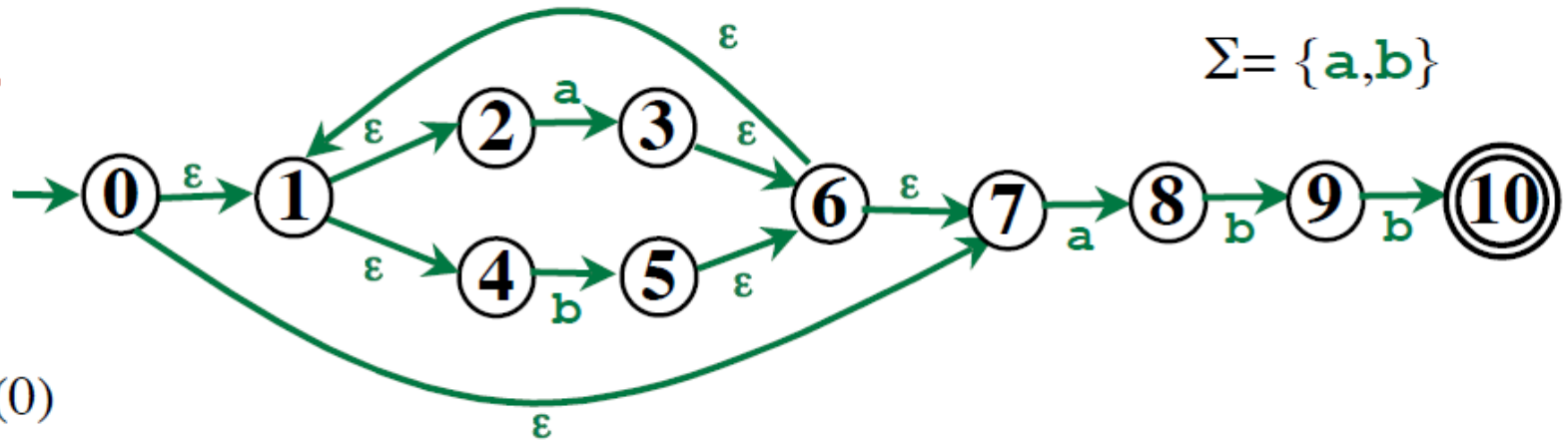
= ε-Closure ({3,8})

=

$Move_{DFA}(A,b)$

=

Example



$\Sigma = \{a,b\}$

Start state:

ε-Closure (0)

= {0, 1, 2, 4, 7} = A

$Move_{DFA}(A,a)$

= ε-Closure ($Move_{NFA}(A,a)$)

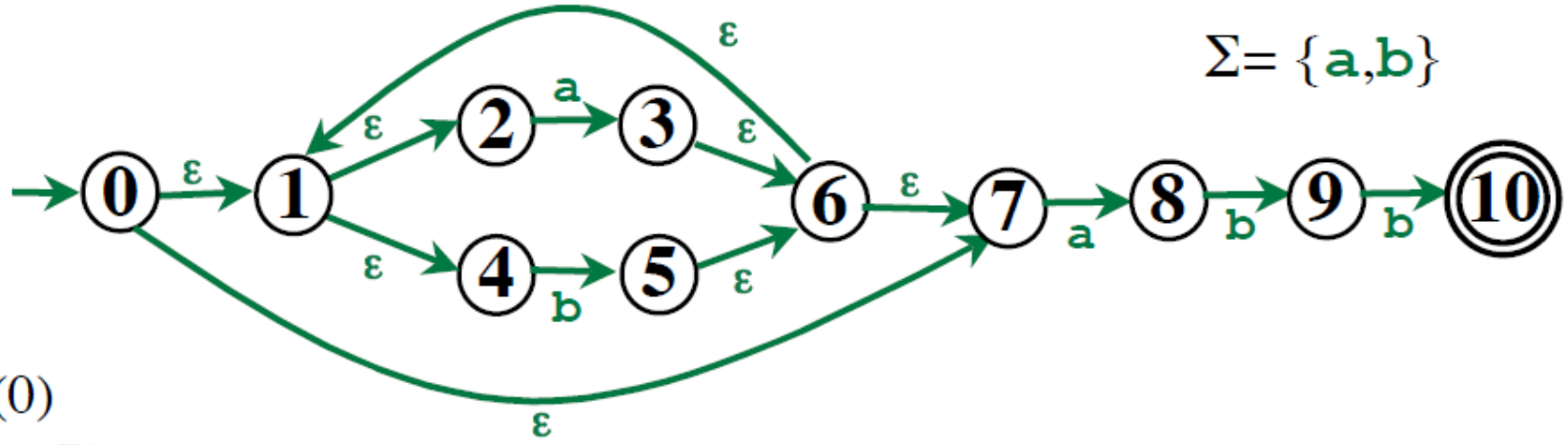
= ε-Closure ({3,8})

= {1,2,3,4,6,7,8}

$Move_{DFA}(A,b)$

=

Example



$\Sigma = \{a,b\}$

Start state:

ε-Closure (0)

= {0, 1, 2, 4, 7} = A

$Move_{DFA}(A,a)$

= ε-Closure ($Move_{NFA}(A,a)$)

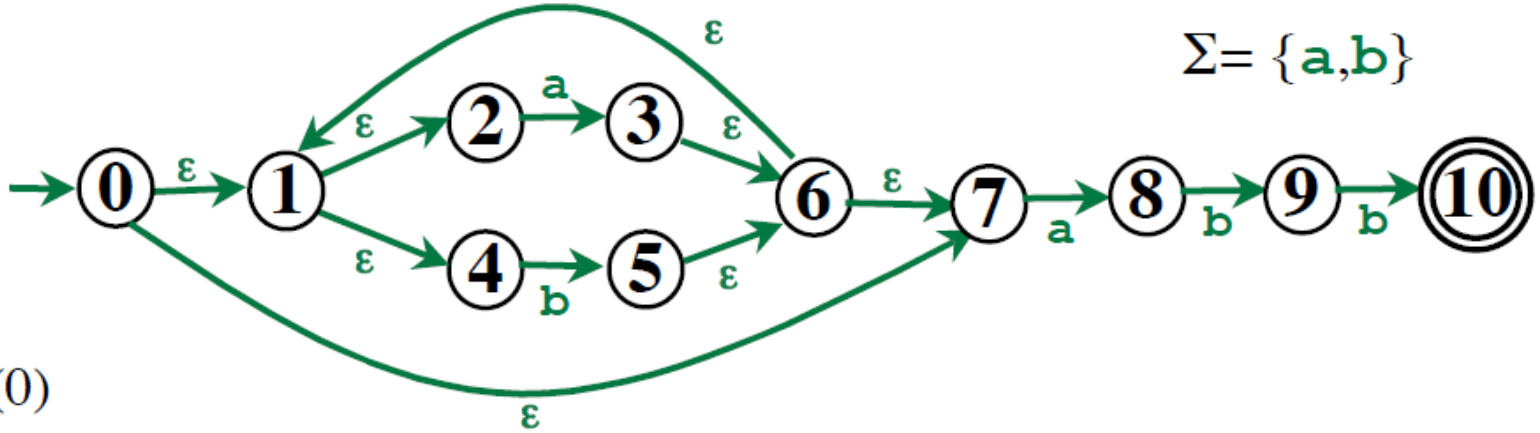
= ε-Closure ({3,8})

= {1,2,3,4,6,7,8} = B

$Move_{DFA}(A,b)$

=

Example



$\Sigma = \{a,b\}$

Start state:

ε-Closure (0)

= {0, 1, 2, 4, 7} = A

$Move_{DFA}(A,a)$

= ε-Closure ($Move_{NFA}(A,a)$)

= ε-Closure ({3,8})

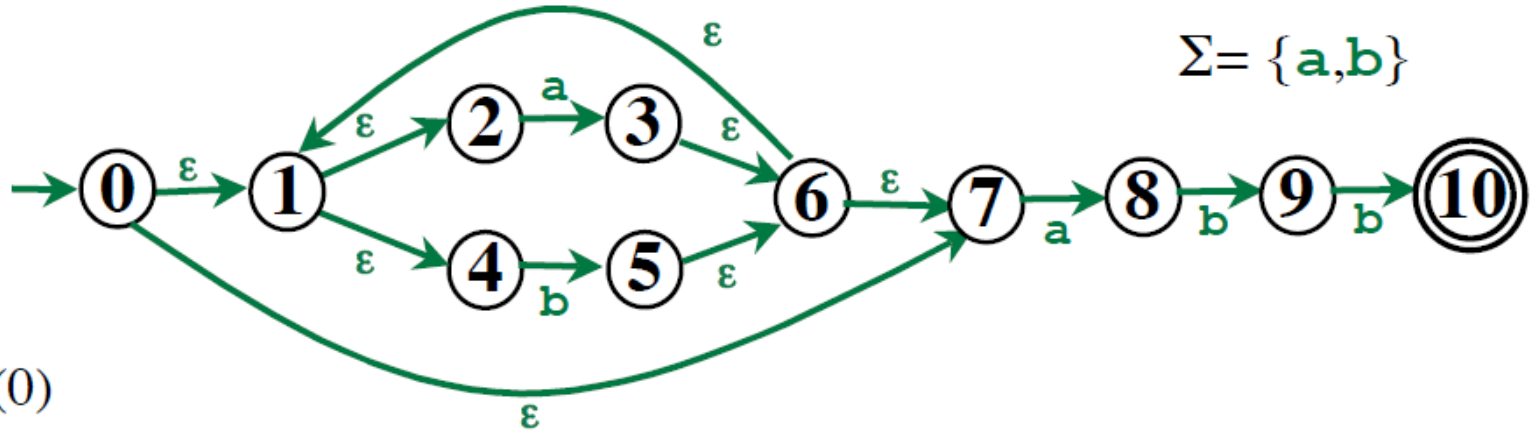
= {1,2,3,4,6,7,8} = B

$Move_{DFA}(A,b)$

= ε-Closure ($Move_{NFA}(A,b)$)

=

Example



$\Sigma = \{a,b\}$

Start state:

ε-Closure (0)

= {0, 1, 2, 4, 7} = A

$Move_{DFA}(A,a)$

= ε-Closure ($Move_{NFA}(A,a)$)

= ε-Closure ({3,8})

= {1,2,3,4,6,7,8} = B

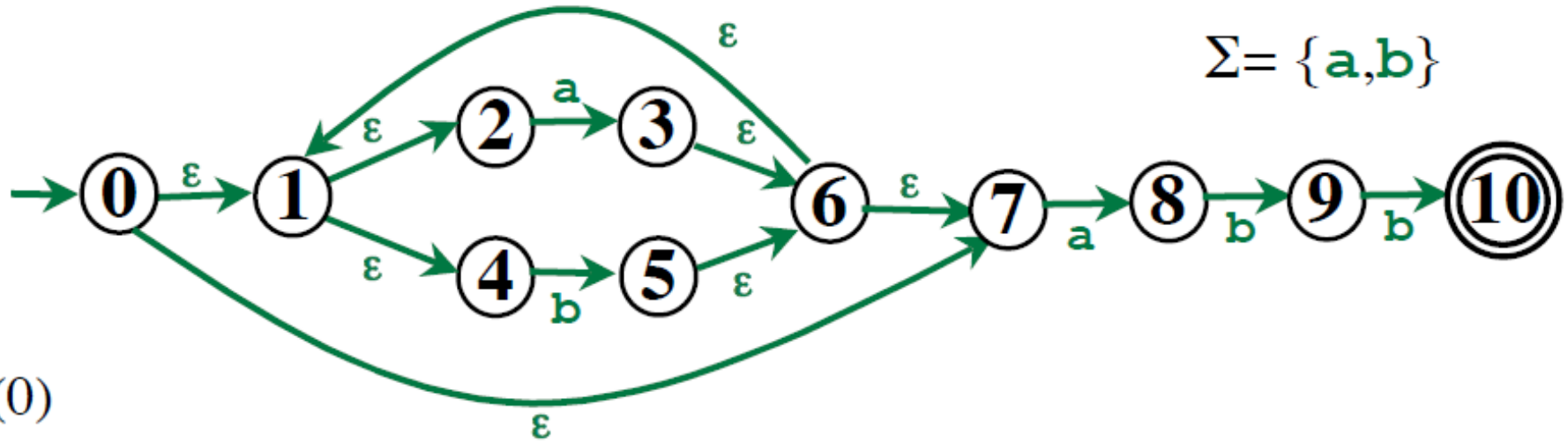
$Move_{DFA}(A,b)$

= ε-Closure ($Move_{NFA}(A,b)$)

= ε-Closure ({5})

=

Example



$\Sigma = \{a,b\}$

Start state:

ε-Closure (0)

= {0, 1, 2, 4, 7} = A

$\text{Move}_{\text{DFA}}(A,a)$

= ε-Closure ($\text{Move}_{\text{NFA}}(A,a)$)

= ε-Closure ({3,8})

= {1,2,3,4,6,7,8} = B

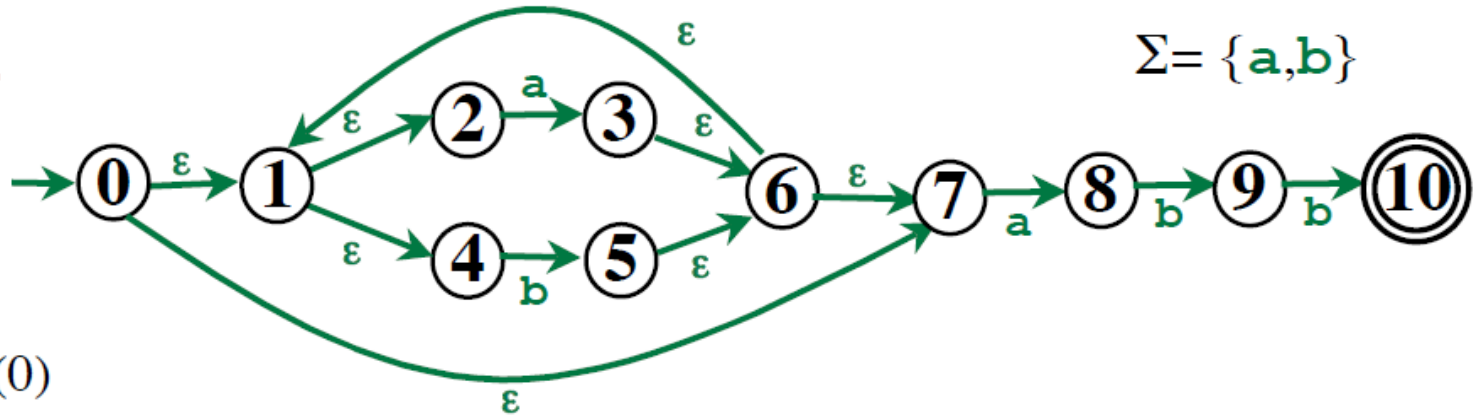
$\text{Move}_{\text{DFA}}(A,b)$

= ε-Closure ($\text{Move}_{\text{NFA}}(A,b)$)

= ε-Closure ({5})

= {1,2,4,5,6,7} = C

Example



$\Sigma = \{a,b\}$

Start state:

ε-Closure (0)

= {0, 1, 2, 4, 7} = A

$\text{Move}_{\text{DFA}}(A,a)$

= ε-Closure ($\text{Move}_{\text{NFA}}(A,a)$)

= ε-Closure ({3,8})

= {1,2,3,4,6,7,8} = B

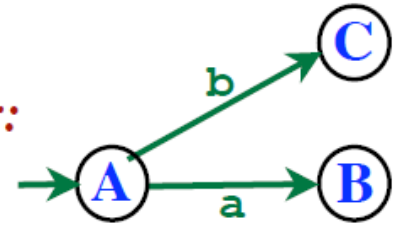
$\text{Move}_{\text{DFA}}(A,b)$

= ε-Closure ($\text{Move}_{\text{NFA}}(A,b)$)

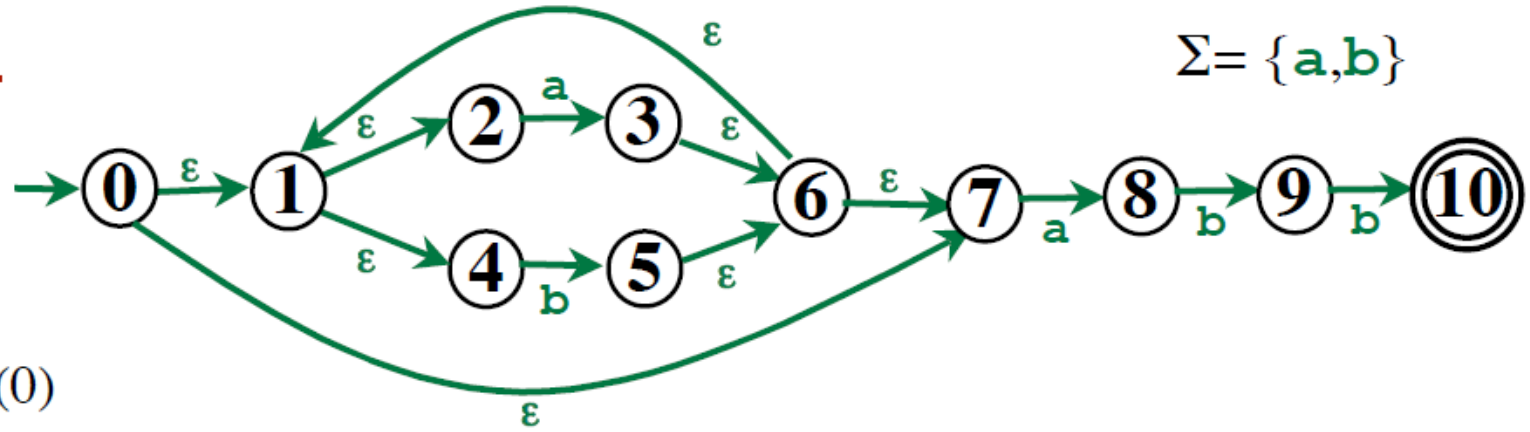
= ε-Closure ({5})

= {1,2,4,5,6,7} = C

So far:



Example



Start state:

ε-Closure (0)

= {0, 1, 2, 4, 7} = A

Move_{DFA}(A,a)

= ε-Closure (Move_{NFA}(A,a))

= ε-Closure ({3,8})

= {1,2,3,4,6,7,8} = B

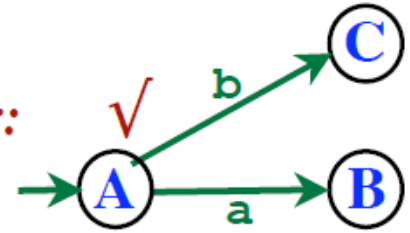
Move_{DFA}(A,b)

= ε-Closure (Move_{NFA}(A,b))

= ε-Closure ({5})

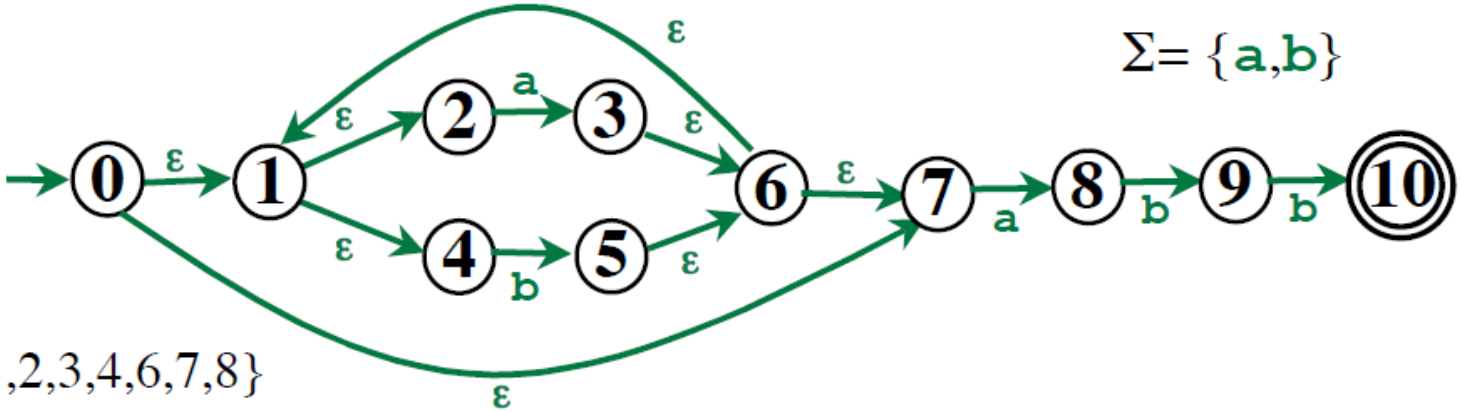
= {1,2,4,5,6,7} = C

So far:



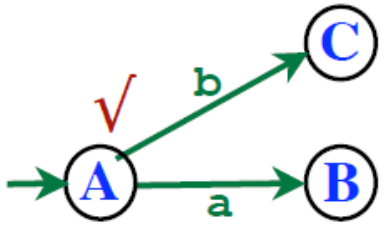
A is now done; mark it!
B and C are unmarked.
Let's do B next...

Example

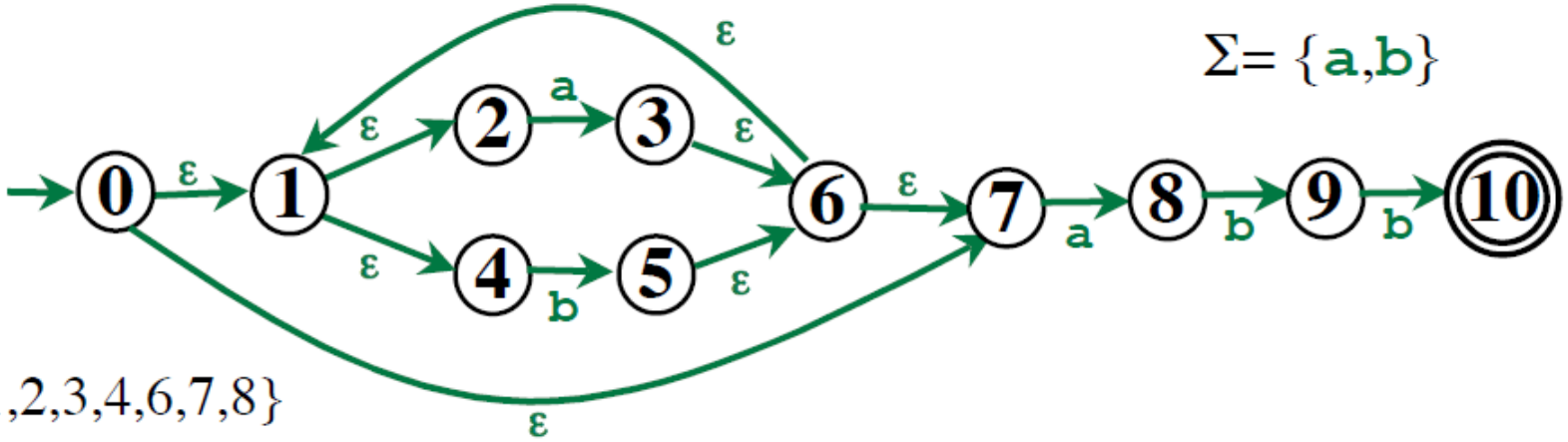


$\Sigma = \{a,b\}$

Process B = {1,2,3,4,6,7,8}



Example

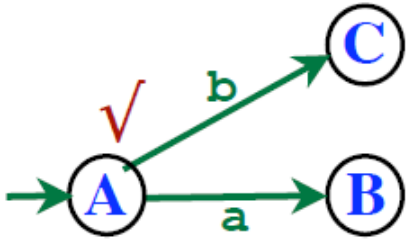


$\Sigma = \{a,b\}$

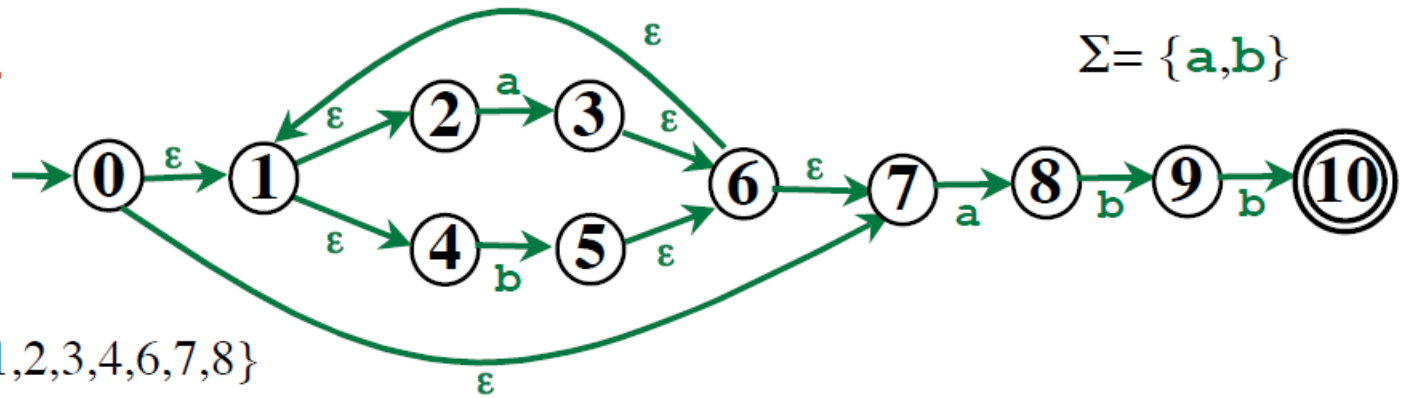
Process B = {1,2,3,4,6,7,8}

$Move_{DFA}(B,a)$

=



Example



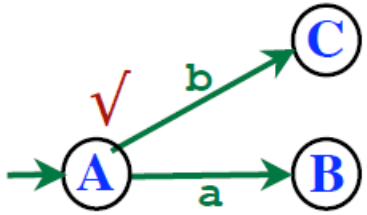
$\Sigma = \{a,b\}$

Process B = {1,2,3,4,6,7,8}

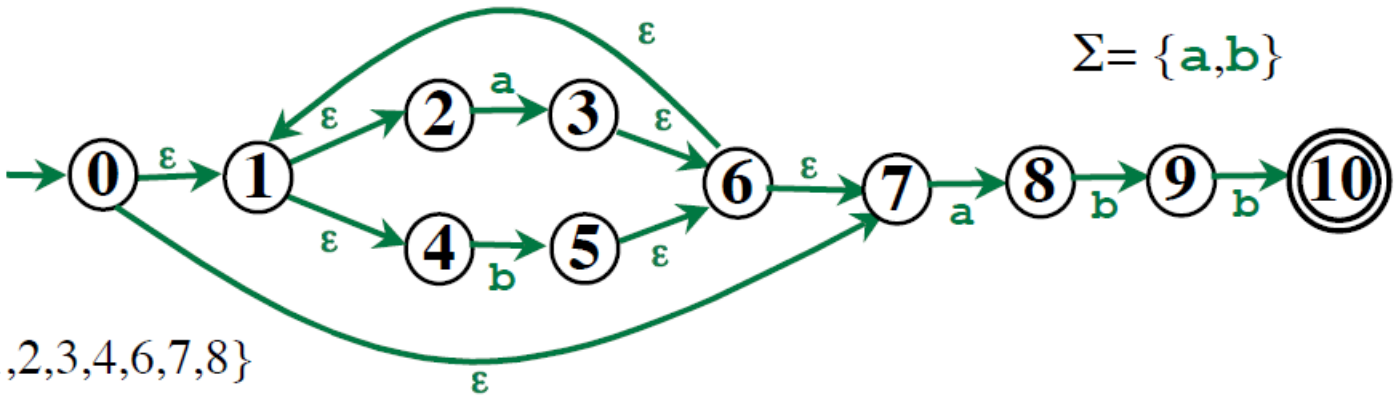
$Move_{DFA}(B,a)$

= ε-Closure ($Move_{NFA}(B,a)$)

=



Example



$\Sigma = \{a,b\}$

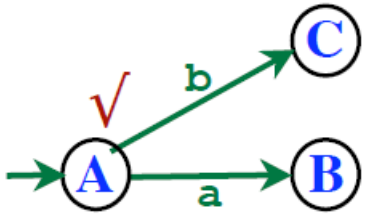
Process B = {1,2,3,4,6,7,8}

$Move_{DFA}(B,a)$

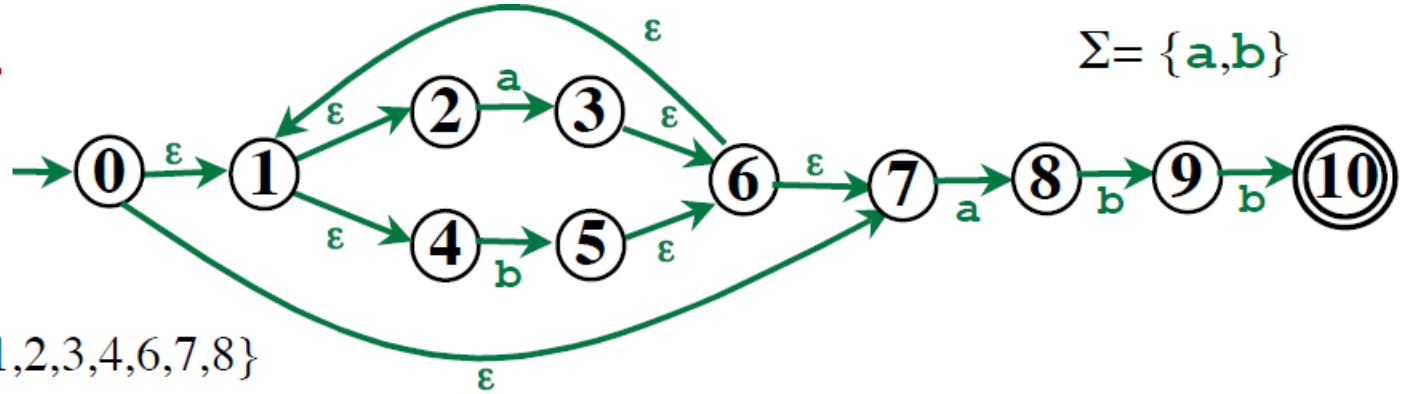
= ε-Closure ($Move_{NFA}(B,a)$)

= ε-Closure ({3,8})

=



Example



$\Sigma = \{a,b\}$

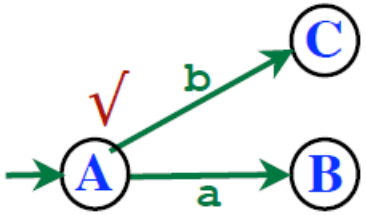
Process B = {1,2,3,4,6,7,8}

$Move_{DFA}(B,a)$

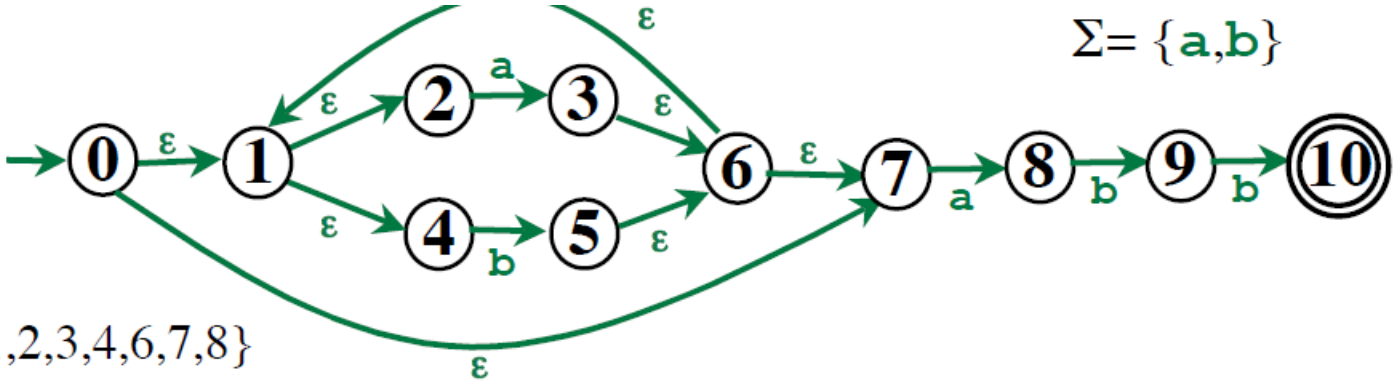
= ε-Closure ($Move_{NFA}(B,a)$)

= ε-Closure ({3,8})

= {1,2,3,4,6,7,8} = B



Example



$\Sigma = \{a,b\}$

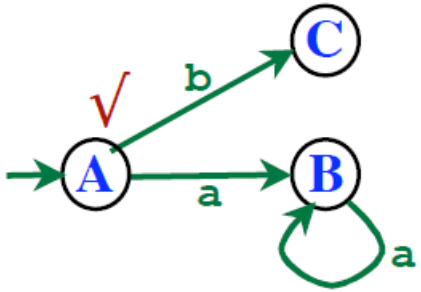
Process B = {1,2,3,4,6,7,8}

$Move_{DFA}(B,a)$

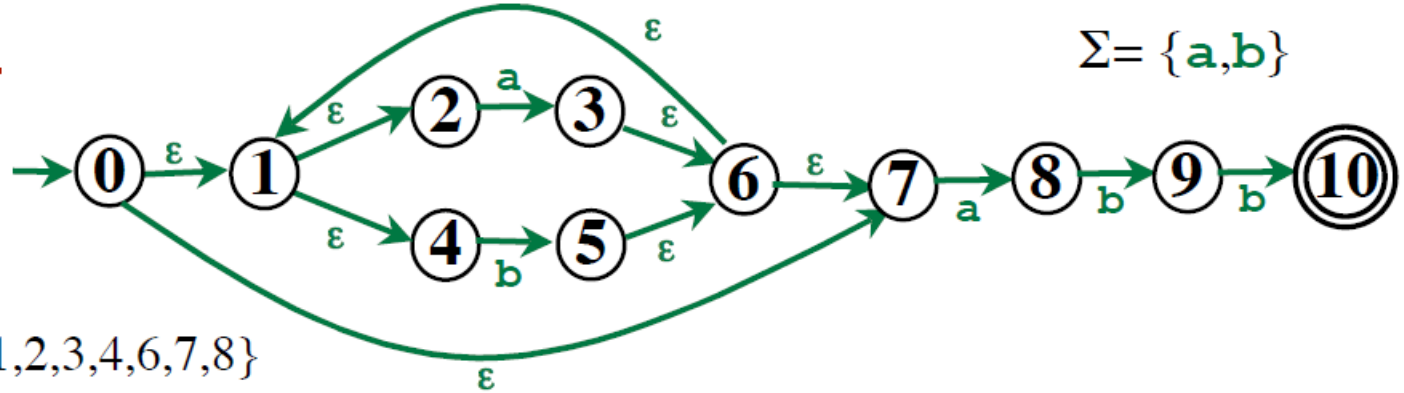
= ε-Closure ($Move_{NFA}(B,a)$)

= ε-Closure ({3,8})

= {1,2,3,4,6,7,8} = B



Example



$\Sigma = \{a,b\}$

Process B = {1,2,3,4,6,7,8}

$Move_{DFA}(B,a)$

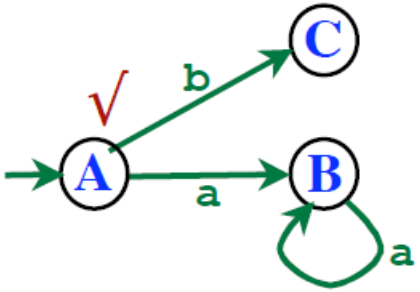
= ε-Closure ($Move_{NFA}(B,a)$)

= ε-Closure ({3,8})

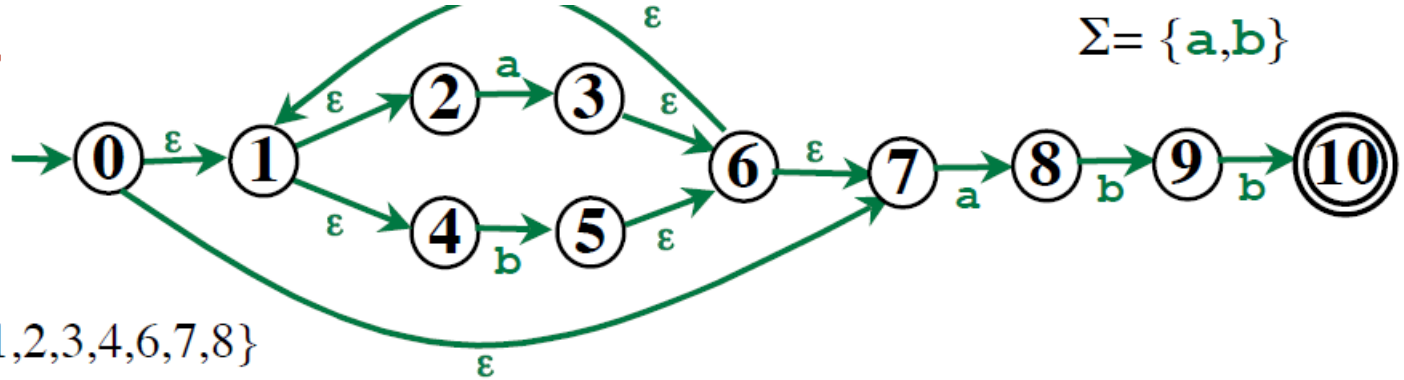
= {1,2,3,4,6,7,8} = B

$Move_{DFA}(B,b)$

=



Example



$\Sigma = \{a,b\}$

Process B = {1,2,3,4,6,7,8}

$\text{Move}_{\text{DFA}}(B,a)$

= ε-Closure ($\text{Move}_{\text{NFA}}(B,a)$)

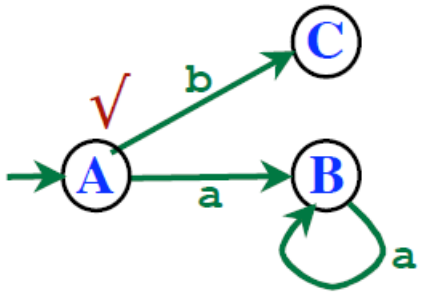
= ε-Closure ({3,8})

= {1,2,3,4,6,7,8} = B

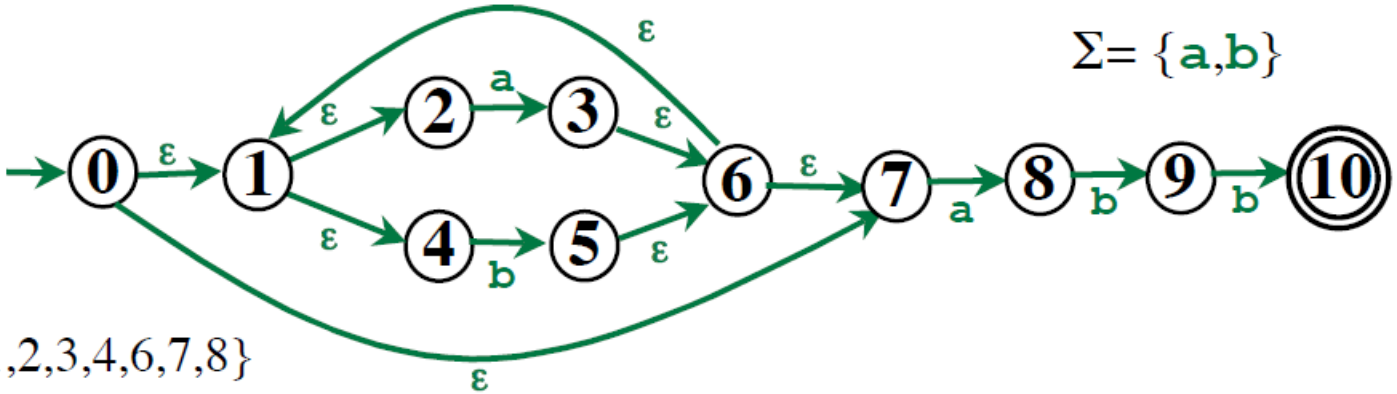
$\text{Move}_{\text{DFA}}(B,b)$

= ε-Closure ($\text{Move}_{\text{NFA}}(B,b)$)

=



Example



$\Sigma = \{a,b\}$

Process B = {1,2,3,4,6,7,8}

$Move_{DFA}(B,a)$

= ε-Closure ($Move_{NFA}(B,a)$)

= ε-Closure ({3,8})

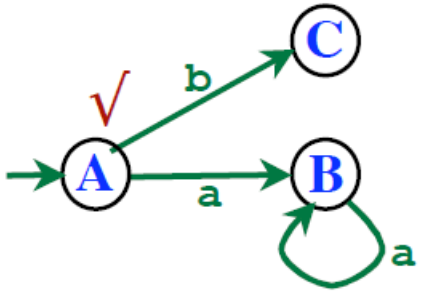
= {1,2,3,4,6,7,8} = B

$Move_{DFA}(B,b)$

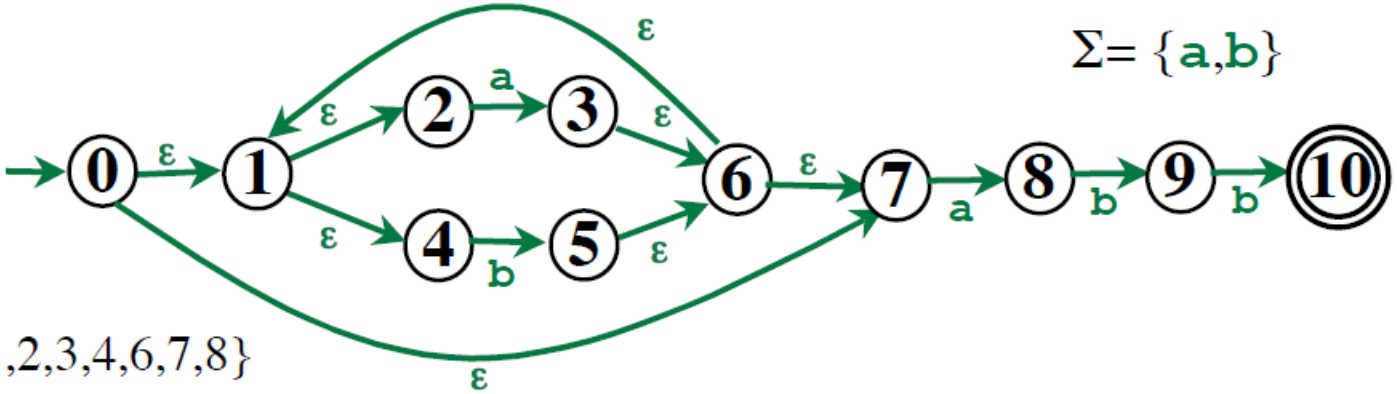
= ε-Closure ($Move_{NFA}(B,b)$)

= ε-Closure ({5,9})

=



Example



$\Sigma = \{a,b\}$

Process B = {1,2,3,4,6,7,8}

$Move_{DFA}(B,a)$

= ε-Closure ($Move_{NFA}(B,a)$)

= ε-Closure ({3,8})

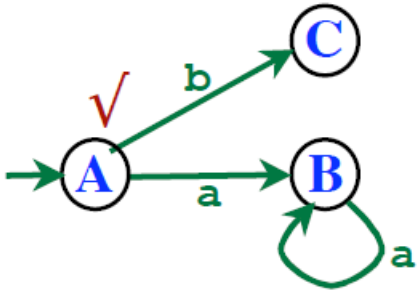
= {1,2,3,4,6,7,8} = B

$Move_{DFA}(B,b)$

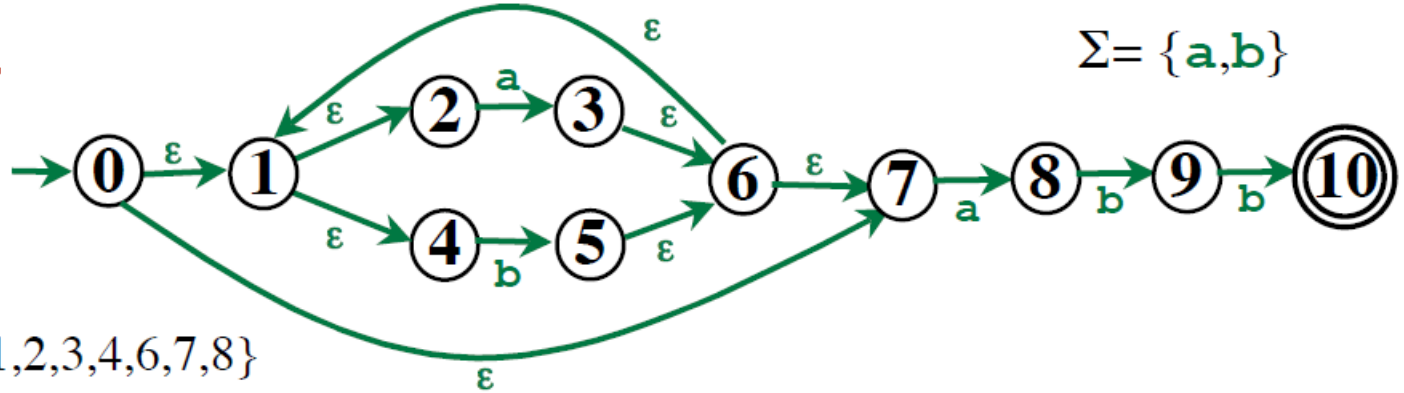
= ε-Closure ($Move_{NFA}(B,b)$)

= ε-Closure ({5,9})

= {1,2,4,5,6,7,9} = D



Example



$\Sigma = \{a,b\}$

Process B = {1,2,3,4,6,7,8}

$Move_{DFA}(B,a)$

= ε-Closure ($Move_{NFA}(B,a)$)

= ε-Closure ({3,8})

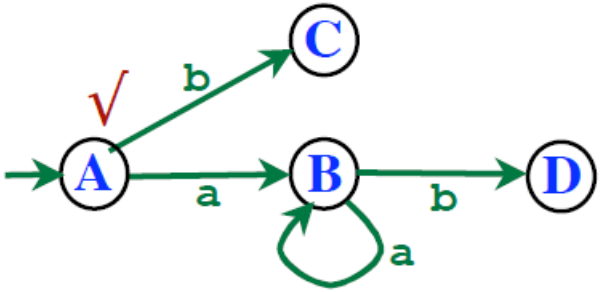
= {1,2,3,4,6,7,8} = B

$Move_{DFA}(B,b)$

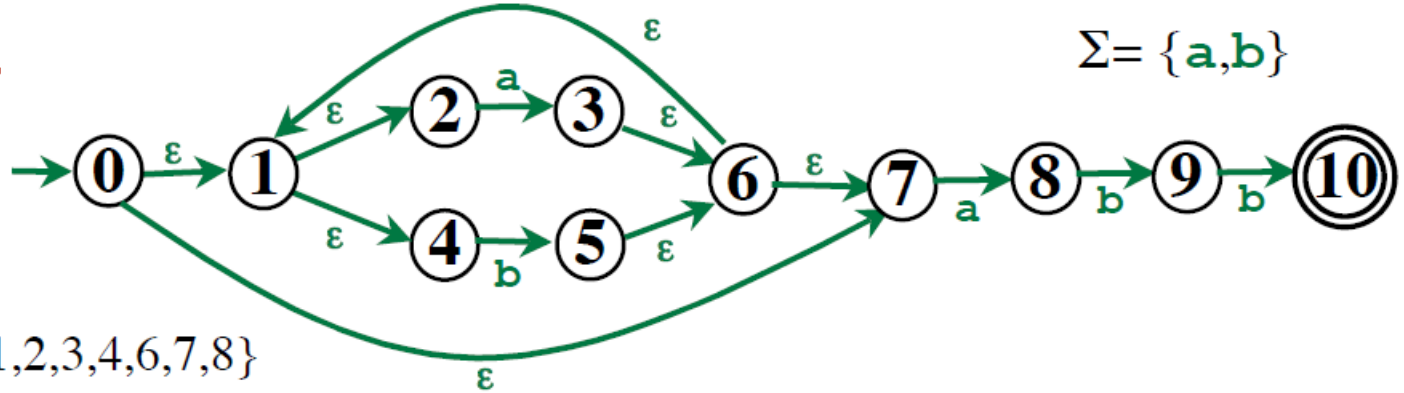
= ε-Closure ($Move_{NFA}(B,b)$)

= ε-Closure ({5,9})

= {1,2,4,5,6,7,9} = D



Example



$\Sigma = \{a,b\}$

Process B = {1,2,3,4,6,7,8}

$Move_{DFA}(B,a)$

= ε-Closure ($Move_{NFA}(B,a)$)

= ε-Closure ({3,8})

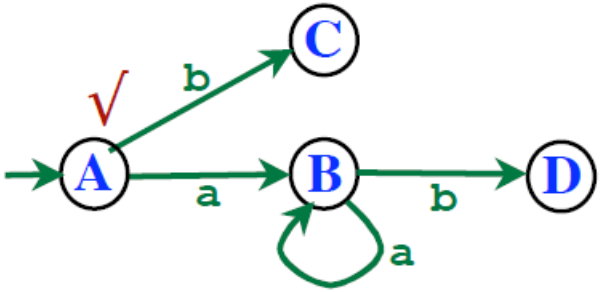
= {1,2,3,4,6,7,8} = B

$Move_{DFA}(B,b)$

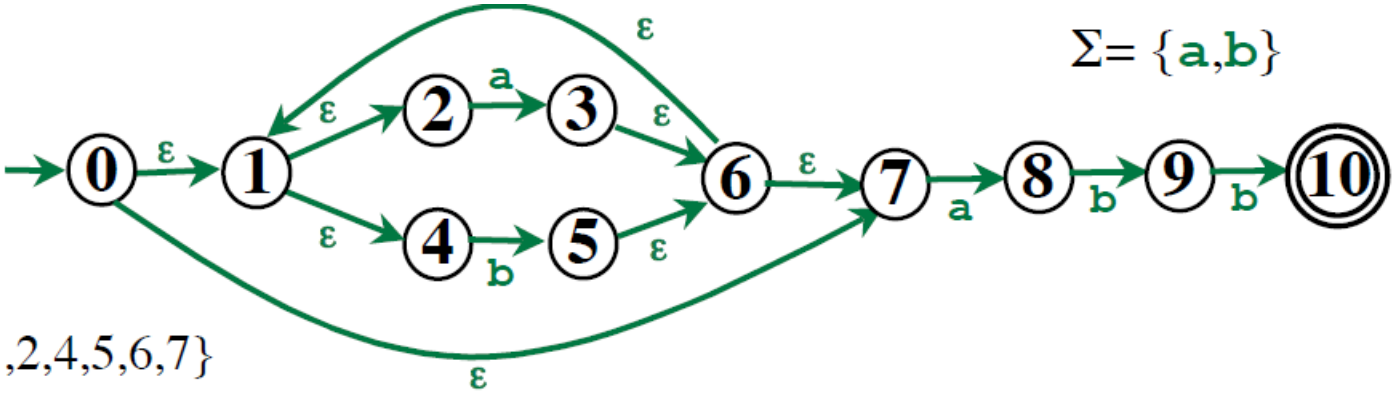
= ε-Closure ($Move_{NFA}(B,b)$)

= ε-Closure ({5,9})

= {1,2,4,5,6,7,9} = D

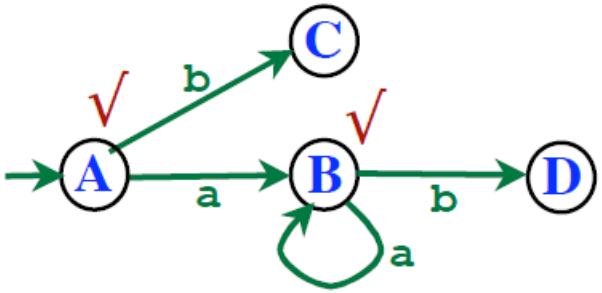


Example

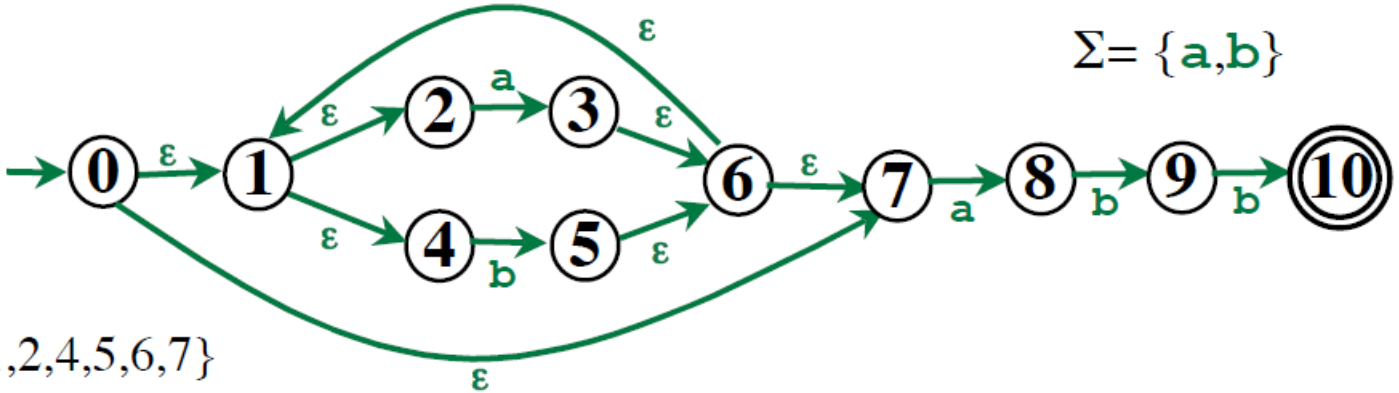


$\Sigma = \{a,b\}$

Process C = {1,2,4,5,6,7}



Example

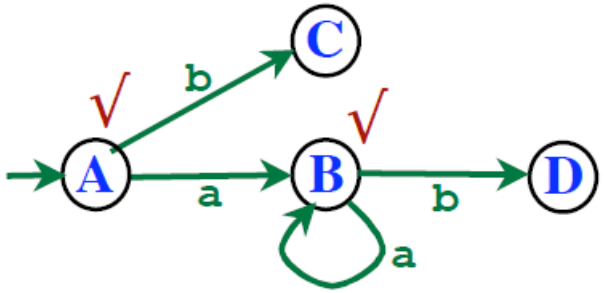


$\Sigma = \{a,b\}$

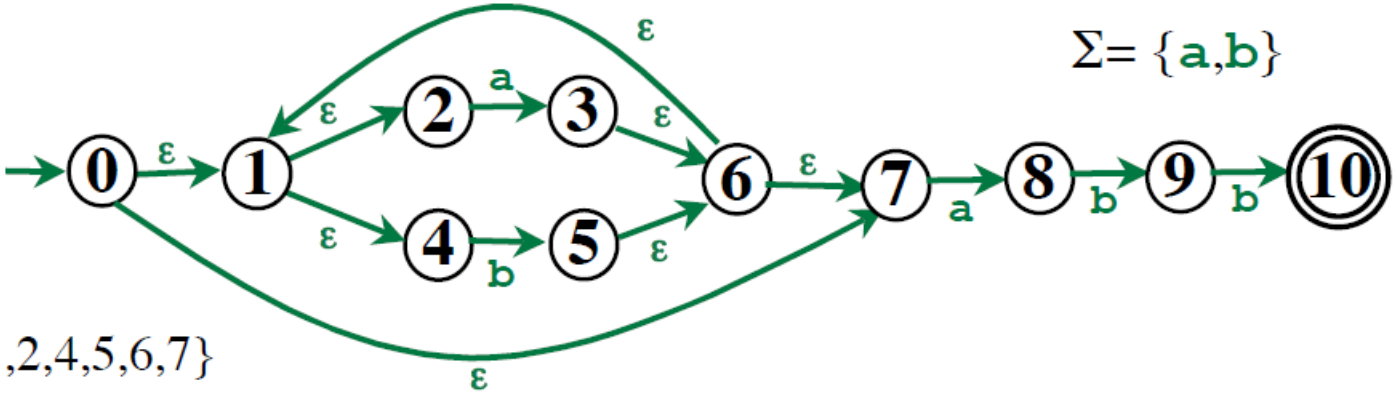
Process C = {1,2,4,5,6,7}

$Move_{DFA}(C,a) =$

$Move_{DFA}(C,b) =$



Example

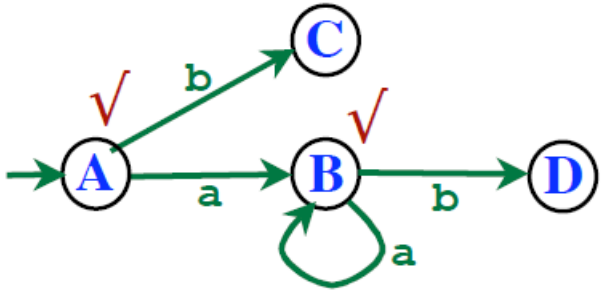


$\Sigma = \{a,b\}$

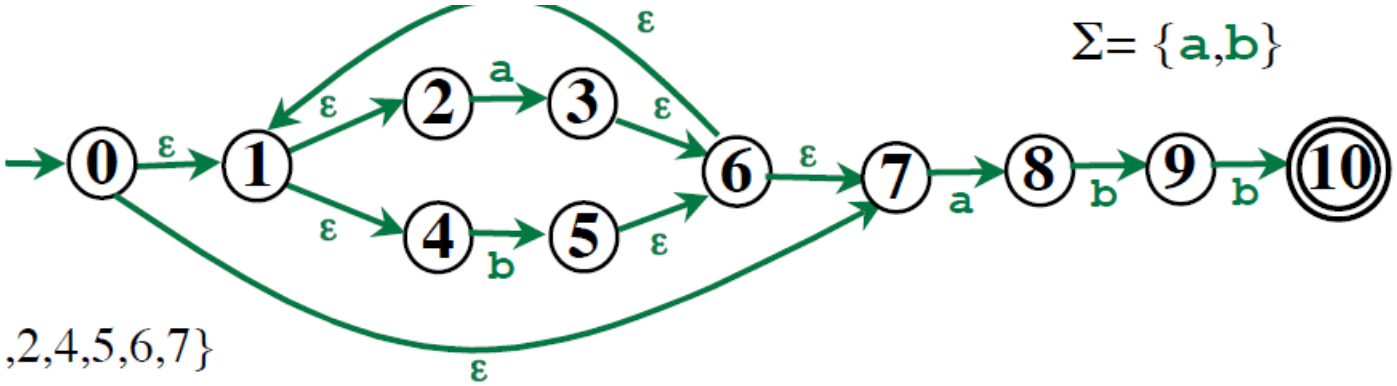
Process C = {1,2,4,5,6,7}

$Move_{DFA}(C,a) = \{1,2,3,4,6,7,8\} = B$

$Move_{DFA}(C,b) =$



Example

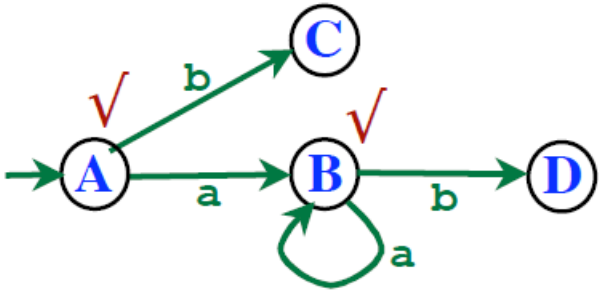


$\Sigma = \{a,b\}$

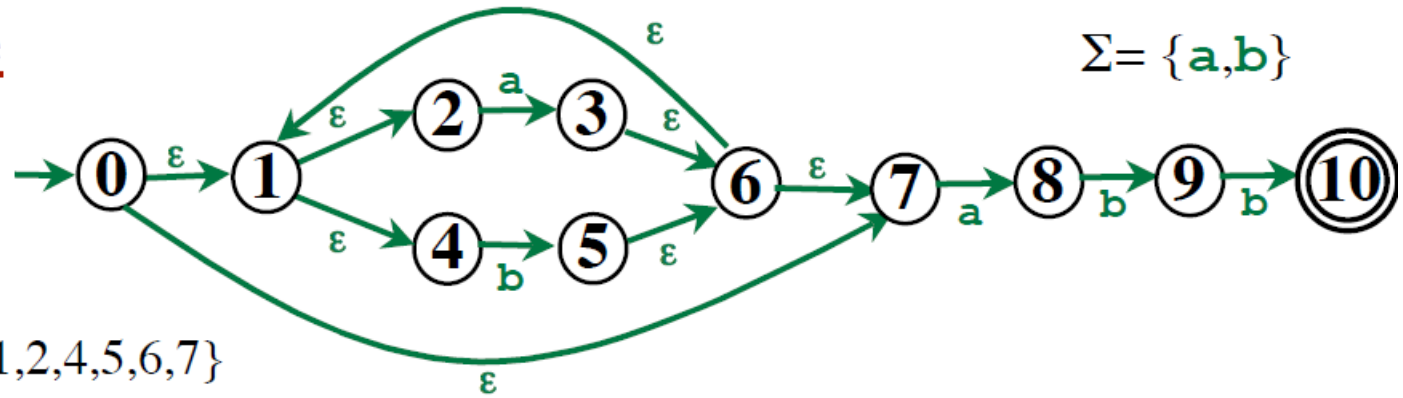
Process C = {1,2,4,5,6,7}

$Move_{DFA}(C,a) = \{1,2,3,4,6,7,8\} = B$

$Move_{DFA}(C,b) = \{1,2,4,5,6,7\} = C$



Example

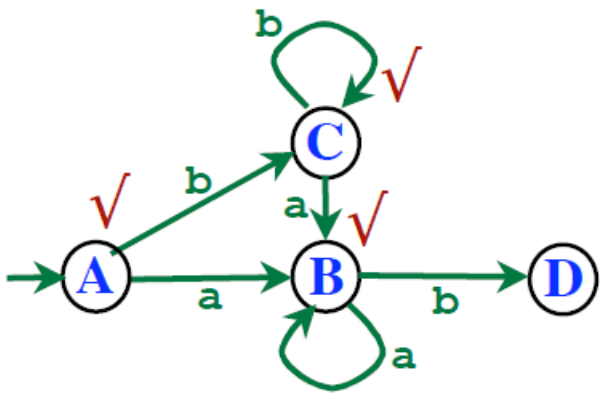


$\Sigma = \{a,b\}$

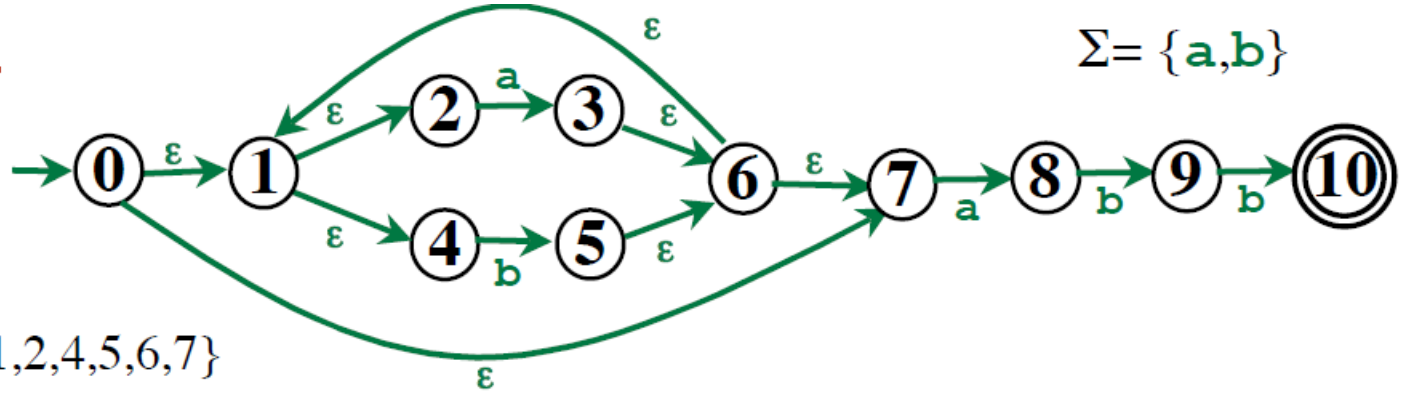
Process C = {1,2,4,5,6,7}

$Move_{DFA}(C,a) = \{1,2,3,4,6,7,8\} = B$

$Move_{DFA}(C,b) = \{1,2,4,5,6,7\} = C$



Example



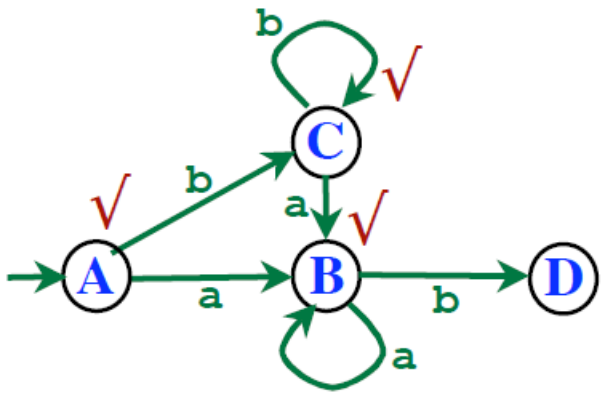
$\Sigma = \{a,b\}$

Process C = {1,2,4,5,6,7}

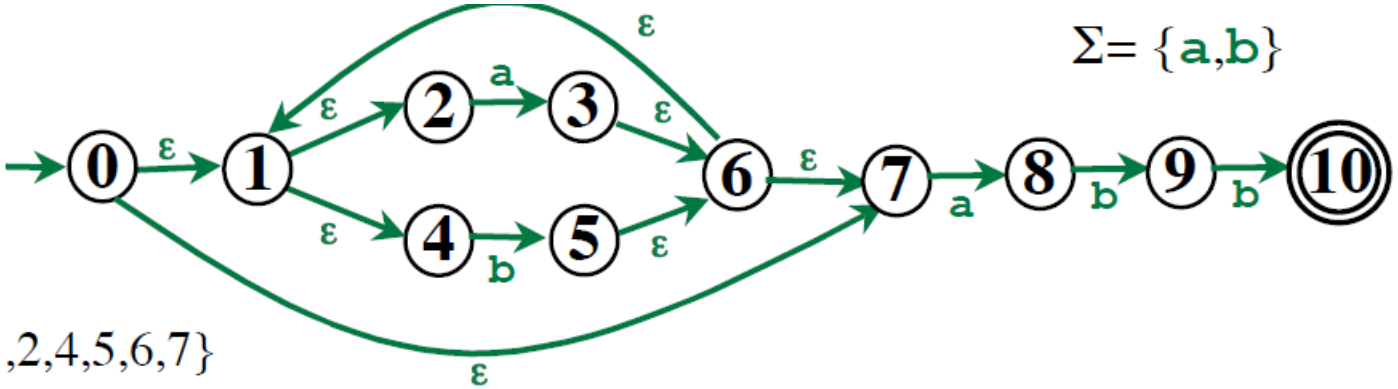
$Move_{DFA}(C,a) = \{1,2,3,4,6,7,8\} = B$
$Move_{DFA}(C,b) = \{1,2,4,5,6,7\} = C$

Process D = {1,2,4,5,6,7,9}

$Move_{DFA}(D,a) =$
$Move_{DFA}(D,b) =$



Example



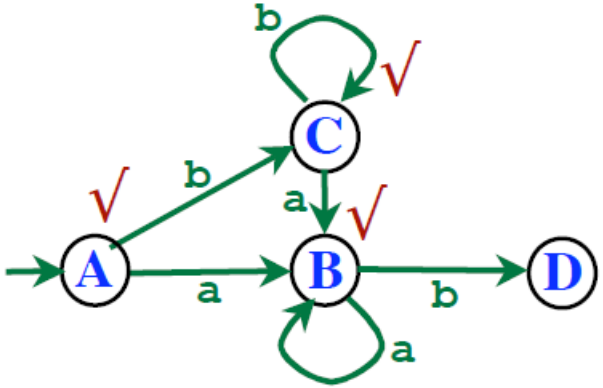
$\Sigma = \{a,b\}$

Process C = {1,2,4,5,6,7}

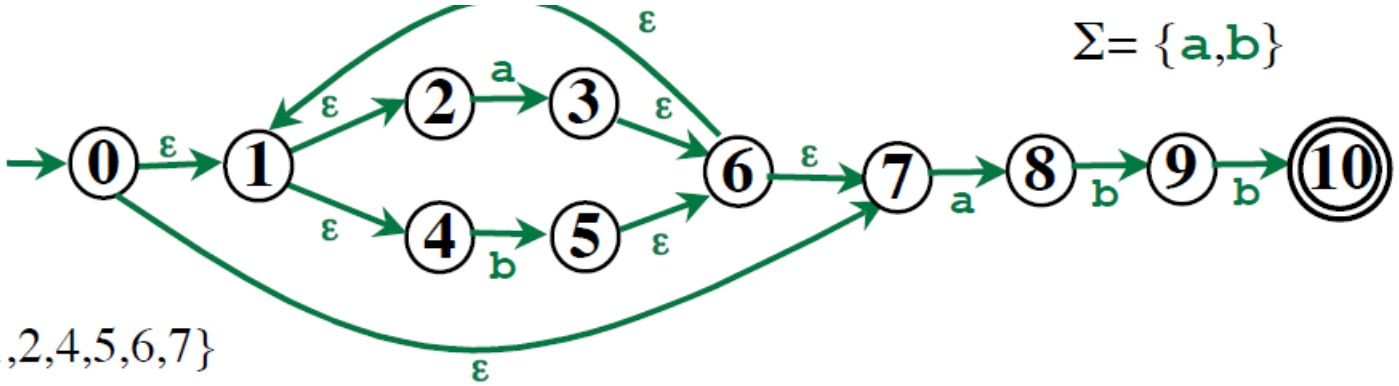
$Move_{DFA}(C,a) = \{1,2,3,4,6,7,8\} = B$
$Move_{DFA}(C,b) = \{1,2,4,5,6,7\} = C$

Process D = {1,2,4,5,6,7,9}

$Move_{DFA}(D,a) = \{1,2,3,4,6,7,8\} = B$
$Move_{DFA}(D,b) =$



Example



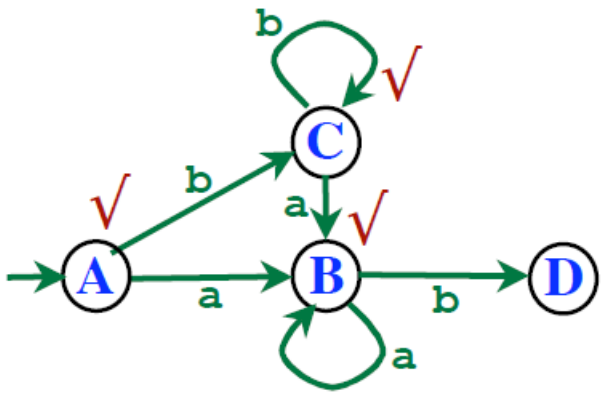
$\Sigma = \{a,b\}$

Process C = {1,2,4,5,6,7}

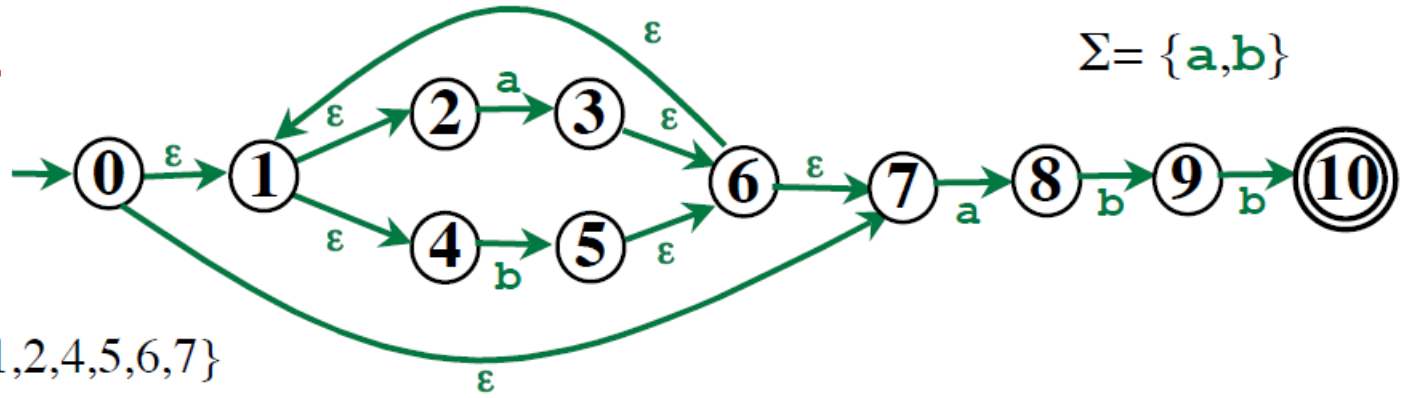
$Move_{DFA}(C,a) = \{1,2,3,4,6,7,8\} = B$
$Move_{DFA}(C,b) = \{1,2,4,5,6,7\} = C$

Process D = {1,2,4,5,6,7,9}

$Move_{DFA}(D,a) = \{1,2,3,4,6,7,8\} = B$
$Move_{DFA}(D,b) = \{1,2,4,5,6,7,10\} = E$



Example



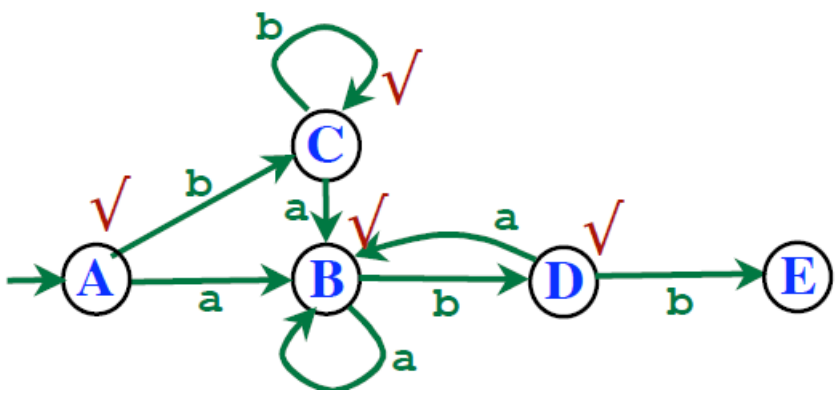
$\Sigma = \{a,b\}$

Process C = {1,2,4,5,6,7}

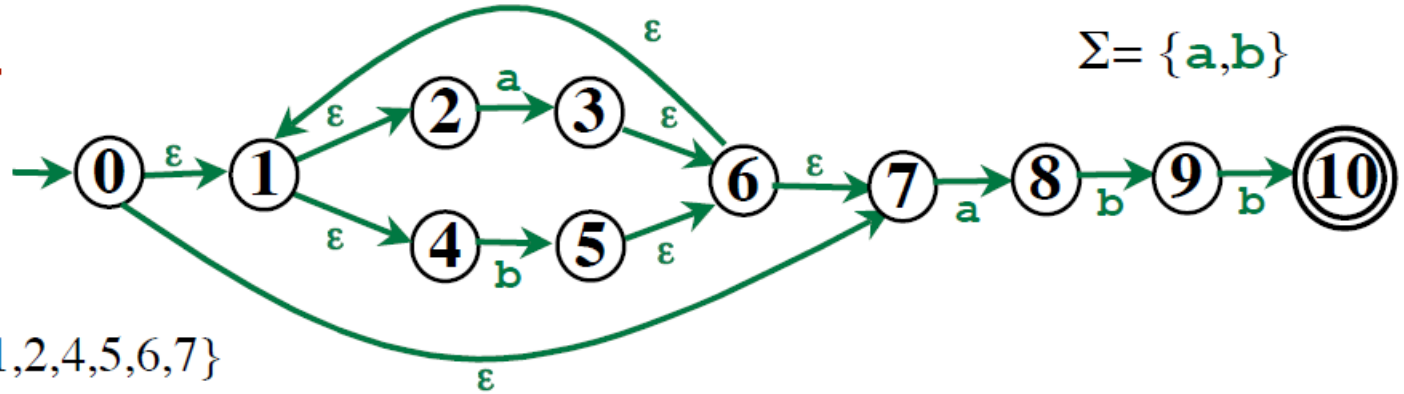
$Move_{DFA}(C,a) = \{1,2,3,4,6,7,8\} = B$
$Move_{DFA}(C,b) = \{1,2,4,5,6,7\} = C$

Process D = {1,2,4,5,6,7,9}

$Move_{DFA}(D,a) = \{1,2,3,4,6,7,8\} = B$
$Move_{DFA}(D,b) = \{1,2,4,5,6,7,10\} = E$



Example



$\Sigma = \{a,b\}$

Process C = {1,2,4,5,6,7}

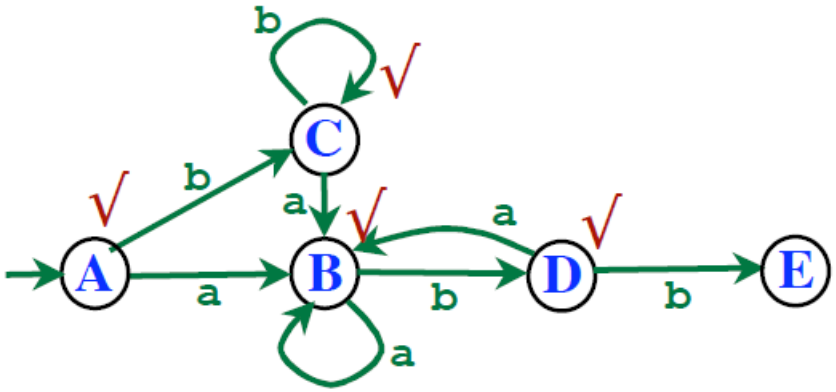
$Move_{DFA}(C,a) = \{1,2,3,4,6,7,8\} = B$
$Move_{DFA}(C,b) = \{1,2,4,5,6,7\} = C$

Process D = {1,2,4,5,6,7,9}

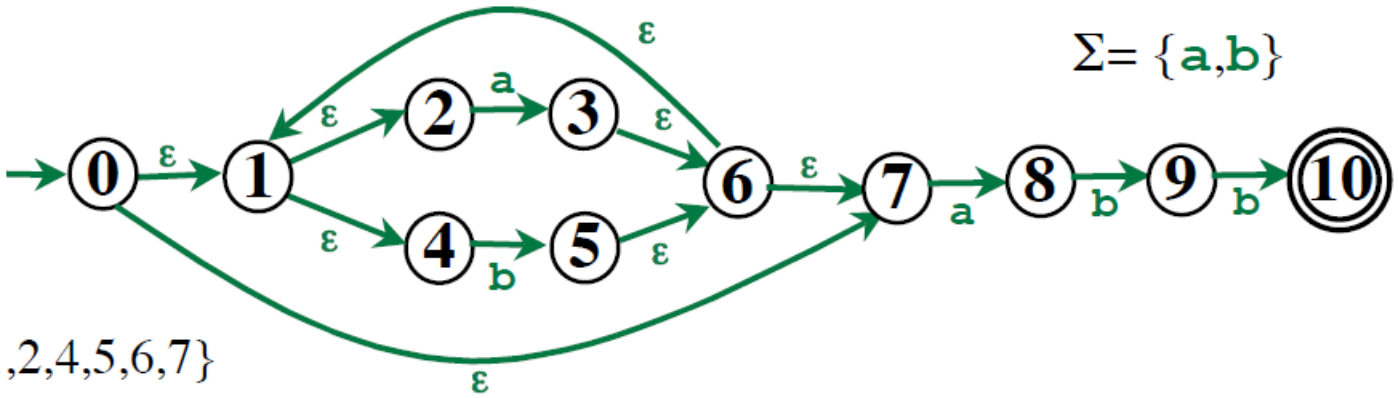
$Move_{DFA}(D,a) = \{1,2,3,4,6,7,8\} = B$
$Move_{DFA}(D,b) = \{1,2,4,5,6,7,10\} = E$

Process E = {1,2,4,5,6,7,10}

$Move_{DFA}(E,a) =$
$Move_{DFA}(E,b) =$



Example



$\Sigma = \{a,b\}$

Process C = {1,2,4,5,6,7}

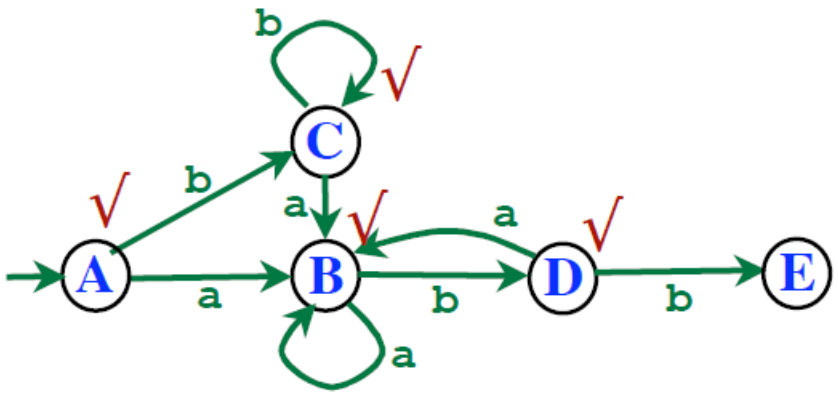
$Move_{DFA}(C,a) = \{1,2,3,4,6,7,8\} = B$
$Move_{DFA}(C,b) = \{1,2,4,5,6,7\} = C$

Process D = {1,2,4,5,6,7,9}

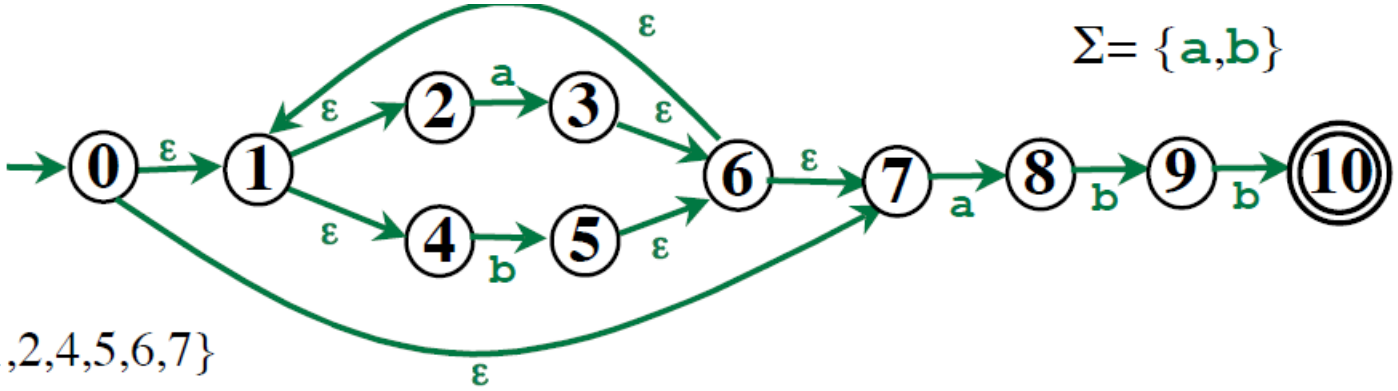
$Move_{DFA}(D,a) = \{1,2,3,4,6,7,8\} = B$
$Move_{DFA}(D,b) = \{1,2,4,5,6,7,10\} = E$

Process E = {1,2,4,5,6,7,10}

$Move_{DFA}(E,a) = \{1,2,3,4,6,7,8\} = B$
$Move_{DFA}(E,b) =$



Example



$\Sigma = \{a,b\}$

Process C = {1,2,4,5,6,7}

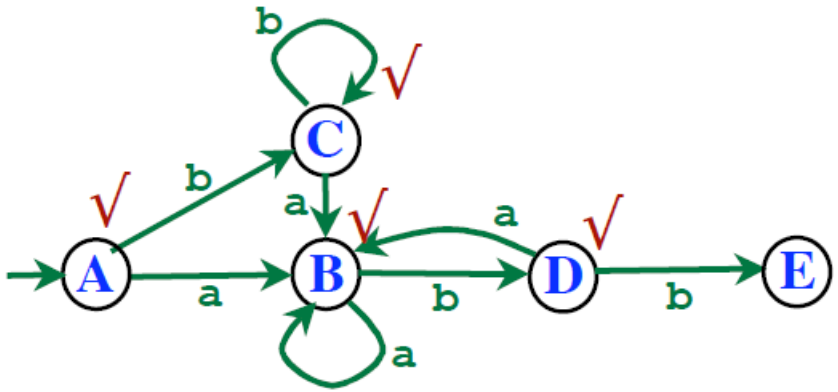
$Move_{DFA}(C,a) = \{1,2,3,4,6,7,8\} = B$
$Move_{DFA}(C,b) = \{1,2,4,5,6,7\} = C$

Process D = {1,2,4,5,6,7,9}

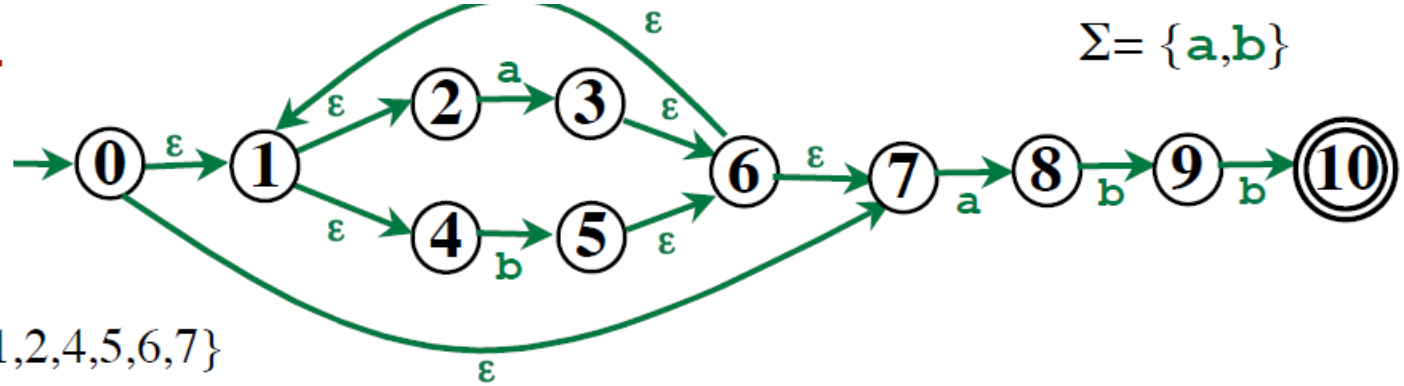
$Move_{DFA}(D,a) = \{1,2,3,4,6,7,8\} = B$
$Move_{DFA}(D,b) = \{1,2,4,5,6,7,10\} = E$

Process E = {1,2,4,5,6,7,10}

$Move_{DFA}(E,a) = \{1,2,3,4,6,7,8\} = B$
$Move_{DFA}(E,b) = \{1,2,4,5,6,7\} = C$



Example



$\Sigma = \{a,b\}$

Process C = {1,2,4,5,6,7}

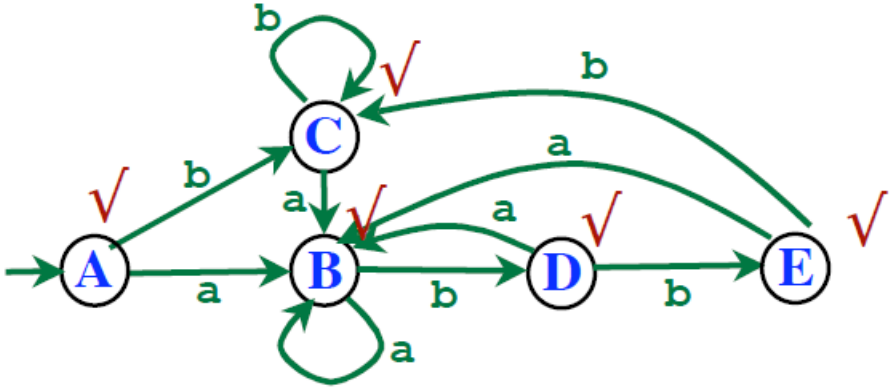
$Move_{DFA}(C,a) = \{1,2,3,4,6,7,8\} = B$
$Move_{DFA}(C,b) = \{1,2,4,5,6,7\} = C$

Process D = {1,2,4,5,6,7,9}

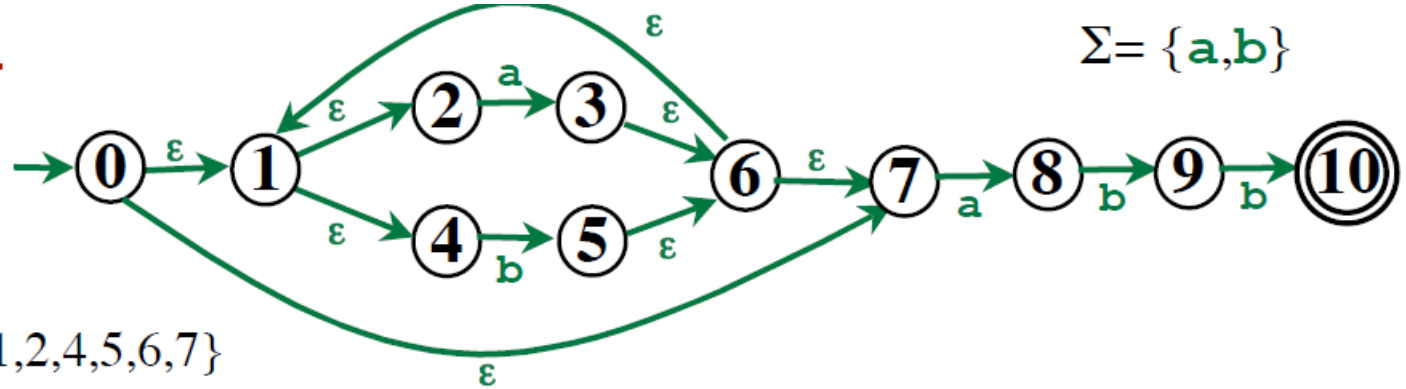
$Move_{DFA}(D,a) = \{1,2,3,4,6,7,8\} = B$
$Move_{DFA}(D,b) = \{1,2,4,5,6,7,10\} = E$

Process E = {1,2,4,5,6,7,10}

$Move_{DFA}(E,a) = \{1,2,3,4,6,7,8\} = B$
$Move_{DFA}(E,b) = \{1,2,4,5,6,7\} = C$



Example



$\Sigma = \{a,b\}$

Process C = {1,2,4,5,6,7}

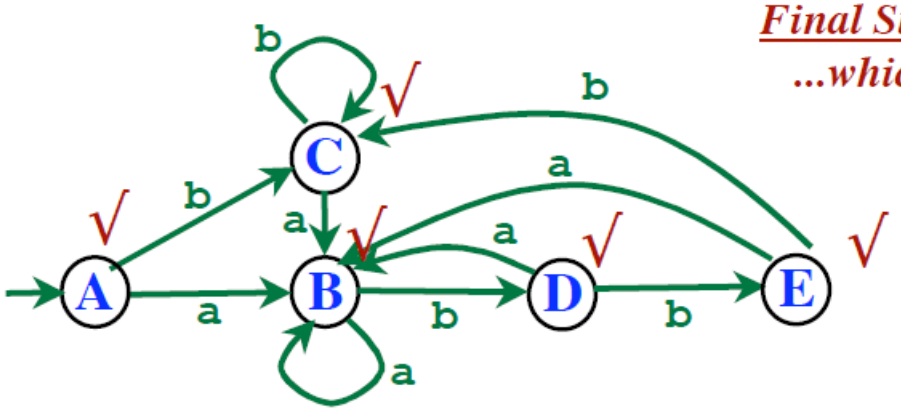
$Move_{DFA}(C,a) = \{1,2,3,4,6,7,8\} = B$
$Move_{DFA}(C,b) = \{1,2,4,5,6,7\} = C$

Process D = {1,2,4,5,6,7,9}

$Move_{DFA}(D,a) = \{1,2,3,4,6,7,8\} = B$
$Move_{DFA}(D,b) = \{1,2,4,5,6,7,10\} = E$

Process E = {1,2,4,5,6,7,10}

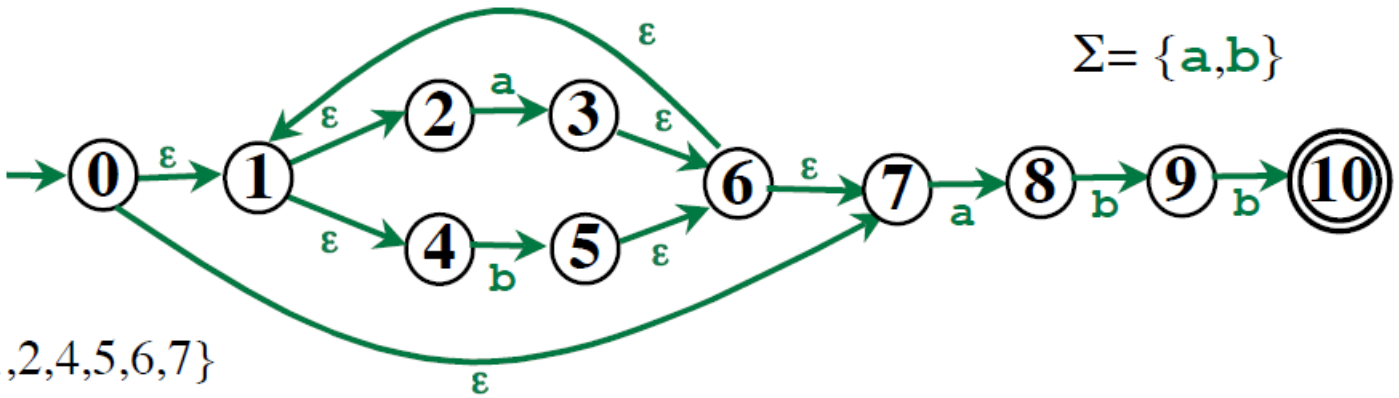
$Move_{DFA}(E,a) = \{1,2,3,4,6,7,8\} = B$
$Move_{DFA}(E,b) = \{1,2,4,5,6,7\} = C$



Final States in DFA?

...which state(s) contain 10?

Example



$\Sigma = \{a,b\}$

Process C = {1,2,4,5,6,7}

$Move_{DFA}(C,a) = \{1,2,3,4,6,7,8\} = B$
$Move_{DFA}(C,b) = \{1,2,4,5,6,7\} = C$

Process D = {1,2,4,5,6,7,9}

$Move_{DFA}(D,a) = \{1,2,3,4,6,7,8\} = B$
$Move_{DFA}(D,b) = \{1,2,4,5,6,7,10\} = E$

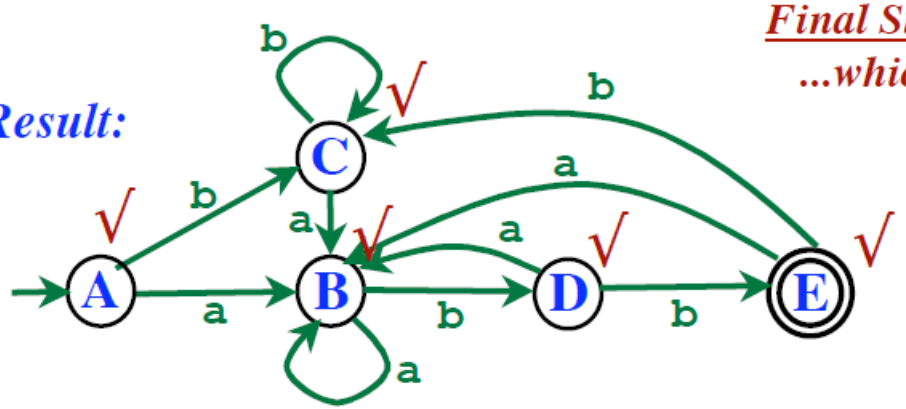
Process E = {1,2,4,5,6,7,10}

$Move_{DFA}(E,a) = \{1,2,3,4,6,7,8\} = B$
$Move_{DFA}(E,b) = \{1,2,4,5,6,7\} = C$

Final States in DFA?

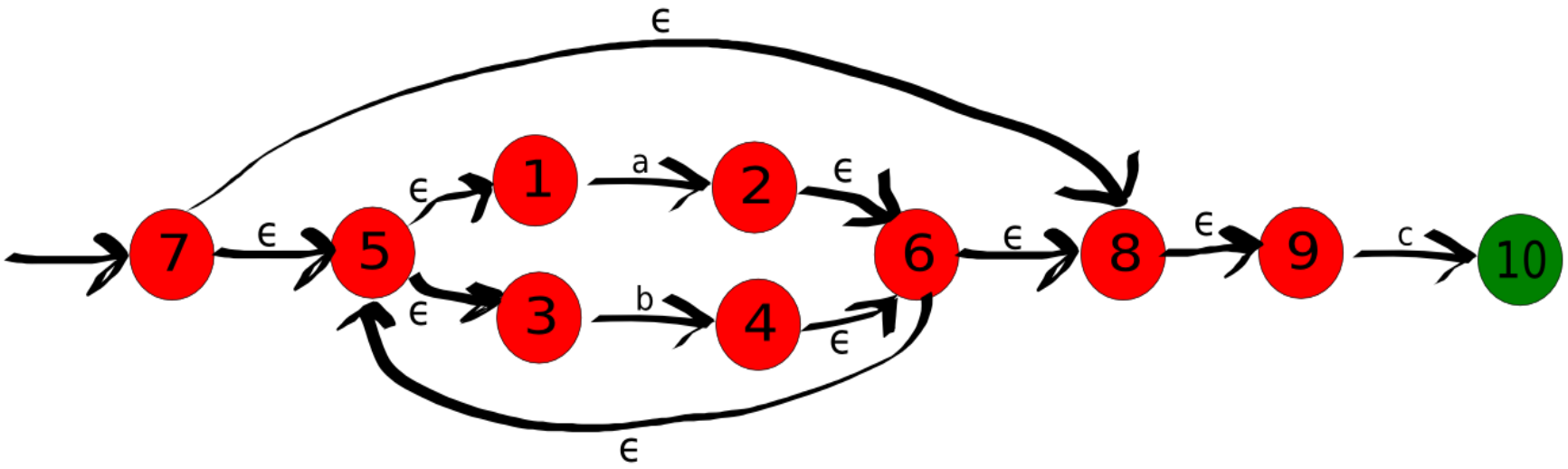
...which state(s) contain 10?

Final Result:



Considering an Example to perform Subset Construction Algorithm to generate DFA from the given NFA

Given NFA



(a|b)*c